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Chapter 1

Introduction

1.1 Background

Cattle monitoring requires us to understand the state of the animal, its behavior and how they change with time. A cow may eat less, rest differently or gradually lose body condition. To be able to fully grasp these changes the observations must also be linked to the particular animal. This points out the importance for indicators taken directly from the animal, and for information on housing and management in Animal welfare studies [3].

Body Condition Scoring (BCS), behavior detection, and individual identification are pertinent categories of information. According to the BCS, the body's reserves are estimated by the visual aspect and quantity of tissue that covers critical areas of the body. Rather than solely relying on body dimensions, scoring systems typically evaluate regions such as the tailhead, vertebrae, loin, and pelvis [1], [2]. Behavior recognition encompasses actions such as eating, drinking, standing, lying down, and walking. Repeated observations are associated with the same animal through an individual identification. When combined, these tasks can provide a more comprehensive understanding of an animal than any single activity.

This is a time consuming process requiring a certain level of expertise to gather this information manually. Short observation periods can easily overlook routine welfare events. Sensors and data processing are used in precision livestock farming to improve the frequency and continuity of monitoring. Sani et al. [48] and Lee et al. [47] discuss the difficulty of converting the obtained measurements into usable information for farm decision making. This requires that the observations are reliable and linked to the right animal over time.

Computer vision provides a useful way to observe appearance, posture, and movement without attaching a separate sensor to every animal. Reviews of vision-based livestock monitoring describe many applications in dairy farming and other livestock settings, but they also report problems when models are used under changing or less controlled conditions [38], [51]. Therefore, the value of computer vision is not only that it can automate one task. It can also provide several related measurements from the same image or video source.

Deep learning allows these visual features to be learned directly from data. Reusable image encoders such as ResNet and EfficientNet reduce the need to design a separate feature-extraction method for every task [4], [18]. Previous cattle studies have used learned features for body condition assessment, behavior recognition, and individual

matching. However, combining these tasks is more difficult because each task may need different visual information.

Multi-task learning (MTL) trains several tasks with some shared model components and separate task outputs. Sharing can help tasks reuse useful features, but it can also hurt performance if the shared features are not suitable for every task. Research on task grouping and selective sharing therefore treats the relationship between tasks as something that should be tested rather than assumed [5], [22]. This thesis studies that problem for cattle body condition, behavior, and identity.

1.2 Rationale of the Study and Motivation

1.2.1 Task-Appropriate Visual Representations

A common MTL design uses one image encoder and places several prediction heads after it. This is simple and efficient, but the fact that several tasks use the same image does not mean that they need the same visual information. Two tasks may look at the same cow while focusing on very different features. This study is motivated by the need to handle those differences inside one framework.

BCS mainly depends on body shape and visible condition-related anatomy. Behavior recognition depends on posture, movement, and sometimes the nearby environment. Re-identification (Re-ID) depends on stable differences between cows, such as coat markings and body shape. Because of this, a feature that helps one task may distract another. Coat pattern is useful for telling two cows apart, but it should not become a shortcut for predicting body condition. A feeding area may help explain an activity, but a Re-ID model should not recognize a cow only because it usually appears in one location.

This problem is associated with shortcut learning. Shortcut learning is when a model learns an easy correlation, rather than the relationship we actually want it to learn. This problem is described by Geirhos et al. [21] in deep learning and image backgrounds can strongly affect recognition as shown by Xiao et al. [27]. The cow and its surrounding environment are present in farm images, so it is important to understand what information the model uses, especially if the camera, farm or recording condition is changed.

In this thesis, *cattle-centred representation learning* is defined as the organization of the visual input around the animal and the task. Localization finds the target cow. Segmentation isolates the visible cow from the rest of the scene. Body structure can be described using pose or anatomical information, and video can be used to show changes in posture and movement. These cues are seen as possibilities to try, not features that are automatically assumed to help every task. The idea is to figure out what can be shared, what should stay task-specific, and when surrounding context is still helpful.

1.2.2 Preliminary Investigations and Their Limitations

A standard pre-trained visual backbone with multiple prediction heads was implemented during the initial phases of the investigation. The multitask configuration described above performed worse than the single-task configuration in terms of body state and behavior. This implied that merely increasing the number of outputs of

a single shared network was insufficient. This prompted us to meticulously examine the information that each step necessitated.

Data review showed that different images are not necessarily independent. Some of the images may be from the same passage of a cow, adjacent moments of the same video, or synchronized perspectives of a single event. A passage label or temporary tracking ID is not necessarily a true biological cow ID either. These details affect how the train-test split evaluates generalization and helps prevent data leakage. Thus, the study groups the observations using the identity and recording information that is actually available in each dataset.

Adding more model components also does not automatically solve these problems. A component may improve localization but still fail to improve the final task. A representation may preserve posture but remove appearance that is useful for identity matching. For this reason, we use single-task models as reference points before studying how the tasks should be combined in one framework.

1.2.3 Rationale for Jointly Studying the Three Tasks

The three tasks are related in practice, but they need different visual information. BCS focuses on morphology, behavior recognition focuses on activity, and Re-ID focuses on individual appearance. Their connection makes integration useful, while their differences make the amount of sharing an important research question.

These tasks also react differently to change. Identity features should remain useful even when the cow changes activity, while behavior features must change when the activity changes. This is different from combining two very similar outputs, such as disease type and disease severity. Studying BCS, behavior, and Re-ID together allows us to examine where shared features help and where task-specific processing is needed.

Pretrained perception models make these comparisons possible without training every supporting model from the beginning. Promptable segmentation and transferable animal pose estimation can provide foreground and structural information [30], [36]. However, simply using these tools is not the contribution of the thesis. The contribution comes from testing how task-appropriate cattle-centered information affects the three tasks and how that information can be combined.

1.3 Problem Statement

This thesis investigates learning and sharing of visual representations for cattle BCS and behavior recognition and Re-ID without compromising information required by each task. Generic RGB features are a good starting point, but they can also confound useful animal information with background or environmental correlations. Another problem with heavy parameter sharing is that a shared representation must support tasks that may prefer different features.

For BCS, the model ought to acquire knowledge regarding body-condition data, rather than attributes that distinctly recognize the cow or its environment. The model requires spatial and temporal data without the necessity of associating a specific action with a particular place for behavior identification. In Re-ID, the model is expected to preserve an individual's identification regardless of variations in camera perspective, backdrop, or temporal context. Consequently, satisfactory execution

of a single task does not suffice to demonstrate that the same representation is applicable to all three jobs.

Previous works on anatomy-based BCS, temporal behavior recognition, metric-learning Re-ID, and selective MTL provide different pieces of the solution. The study integrates these concepts, exploring how cattle-centered information changes each task, the appropriate level of information sharing, and the models' performance under varying recording conditions.

The primary research question is:

Are cattle-centric visual representations better than generic shared RGB representations in improving the robustness of body condition score, behavior recognition and cow re-identification and allowing a multi-task framework with less negative transfer?

To address this issue, we must analyze not only the task performance but also the visual information provided to the model, the sharing approach and the evaluation criteria.

1.4 Objectives

The primary objective is to develop and assess a multi-task deep learning framework for BCS, behavior recognition, and cow Re-ID using task-specific cattle-centric visual representations.

The first goal is to create evaluation protocols that preserve the relationships within each dataset. When reliable biological IDs are available, biological cow IDs are used. When they are not available, recording, passage, or session groups are used. This provides a safer basis for comparing models.

The secondary goal is to produce RGB single-task baseline models for the three tasks. These models provide a basis for assessment of the effect of changes in representation, and for the characterization of positive or negative transfer after integration.

The third objective is to study cattle-centered visual information, such as localization, foreground masks, anatomical or pose cues, and viewpoint information. Temporal information is also considered for behavior recognition. The goal is to determine which combinations are useful for each task while keeping the system practical to train and evaluate.

The fourth goal is to compare heavy sharing with a task-specific design that keeps some features or processing paths task-specific. This comparison checks to see if specialization can keep shared information useful without hurting individual tasks. The fifth goal is to evaluate robustness to different acquisition conditions and to the existence of similar external datasets. This ties the model assessments into the larger goal of monitoring cattle outside the specific environment in which it was trained.

These three research questions expressed the objectives:

1. How do cattle-centered representations affect BCS, behavior recognition, and Re-ID relative to their RGB single-task references?
2. How can temporal information and task-specific visual cues be incorporated while preserving the different requirements of condition assessment, activity recognition, and identity matching?

3. How does task-conditioned sharing compare with hard sharing in task performance, negative transfer, and robustness under domain change?

1.5 Methodology in Brief

The study uses a comparative experimental design. The data preparation stage first defines the labels and the grouping units used for evaluation. ScienceDB is the primary BCS dataset, CVB and Kaggle Beef are the main behavior sources, and SideViewCows2026 is the main Re-ID dataset. Different datasets are used because the three tasks need different types of observations and annotations.

Single-task RGB models are used as reference models. Their cattle-centered versions use localization and binary-mask guidance, while the behavior model also uses temporal aggregation. Pose, anatomy, and viewpoint cues are studied separately instead of assuming that they will transfer well across datasets. Upstream perception is checked before its output is used in a downstream model.

The multi-task investigation evaluates alternative integration strategies against matched single-task cattle-centered reference models on identical held-out test sets. A monolithic hard-shared multi-task model evaluates full parameter sharing across tasks. A modular task-private multi-task model evaluates architectural specialization using dedicated residual bottleneck adapters. A gradient-projected multi-task optimization control (PCGrad) evaluates conflict-aware shared-gradient projection while keeping the hard-shared architecture and parameter count strictly matched. Task-wise held-out comparisons quantify outcome-level negative transfer across prediction error for BCS, class-sensitive metrics for behavior recognition, and open-set retrieval for cow re-identification. Compatible external datasets provide candidate protocols for future cross-environment stress testing, treated separately from primary benchmark evaluations.

In the chapter 4, it discuss about model settings, training methods, experimental controls, as well as data divisions. And in the chapter 5, it discuss about the result as well as the interpretation of the result. These experiments show what information each task needs and what happens when the tasks learn together.

1.6 Scopes and Challenges

The framework covers three visual tasks, BCS, behavior recognition and individual cow identification (Re-ID). The thesis does not cover lameness detection. RGB imagery is the common visual input. Depth and other external data are only used when their role and label compatibility are clear. The presented framework is a research contribution to the monitoring of cattle and not a replacement of the veterinary assessment.

The tasks and datasets give different type of supervision. BCS labels vary in coverage and scoring. The sort of behavior depends on the position and the activity. certain categories are present in certain data only. Re-identification requires a well defined division between training and assessment datasets, and robust bovine identity data. ScienceDB does not do biological cow-disjoint assessment, but rather burst-group-disjoint and sequence-safe evaluation. These differences restrict the assertions that may be made from each experiment.

Viewpoint, occlusion, scale, and lighting also affect what can be seen in an image. Foreground processing can remove irrelevant background, but it may also remove useful context. Pose estimation can describe body structure, but it may still miss anatomical information needed for BCS. Therefore, the study treats component selection as something that must be tested experimentally.

Computational limits also affect the design. We therefore use pretrained models, reusable preprocessing, and focused comparisons instead of making the architecture unnecessarily large. Chapter 2 reviews the related research and the research gap. Chapter 3 describes the requirements, impacts, and constraints. Chapters 4 and 5 present the methodology and experimental analysis, and Chapter 6 presents the conclusions.

The completed study combines leakage-aware evaluation, matched single-task cattle-centered comparisons, and controlled multi-task experiments. The single-task results show task-dependent benefits from cattle-centered representations, while the multi-task comparisons show outcome-level degradation under hard sharing and selective mitigation under alternative architectural and optimization strategies. These findings motivate a unified framework that preserves shared cattle information while recognizing task-specific representation and optimization requirements.

Chapter 2

Literature Review

2.1 Preliminaries

A unified cattle-monitoring system needs to connect different kinds of visual information. Body condition is mainly related to body shape and morphology, behavior is related to posture and movement, and identity depends on features that make one animal different from another. Because of this, the thesis connects ideas from livestock monitoring, visual representation learning, and multi-task learning. This chapter reviews how earlier methods select and use visual information, and how they are evaluated under different conditions.

The chapter first explains the main concepts needed for the three tasks. It then reviews cattle-specific models, foreground processing, anatomy and pose, temporal information, and task-specific feature sharing. These topics are brought together to explain the research gap of this thesis.

2.1.1 Visual Representations and Transfer Learning

A visual representation is the set of features a model uses to describe an image or video sequence. Older recognition systems often depended on hand-designed measurements. Deep neural networks can instead learn useful features directly from examples. In a convolutional network, simple local patterns are gradually combined into higher-level features. The features passed to the final prediction head decide what information is available for making the prediction.

Reusable image encoders have made this easier in specialized fields. ResNet uses residual connections to make deep networks easier to train [4]. EfficientNet focuses on scaling network depth, width, and input resolution together [18]. These models solve different design problems, but both are useful starting points for transfer learning. Instead of learning every visual feature from a small cattle dataset, a pretrained model can be adapted to the cattle task.

Transfer learning and cattle-centered representation learning are not the same in terms of functionality. Transfer learning focuses on the source of initial weights of the model. Cattle-centered learning gives us a clue as to what kind of input data was fed to the model. The same pre-trained encoder may be used with different types of input data such as full image, crop of the cow, masked cow, and anatomically detailed images. That is why the encoder will stay the same regardless of the changes in the input data. This difference is important when comparing why one

representation performs better than another.

2.1.2 Prediction Tasks and Their Visual Requirements

There are different approaches for predicting BCS, which are classification, ordinal prediction, and regression. In classification, all the scores can be treated as individual classes. In ordinal prediction, order is significant since the prediction of a neighboring score is a less significant error compared to the one which is distant. However, regression just tries to predict the numeric output. CORAL is one example of an algorithm designed for the ordered labels [19]. This algorithm may teach the classifier about the order of BCS, but it will not choose the visual features of the cow.

In the case of behavior recognition, a class is actually an activity that is not measured by numbers or scores. One frame can be able to inform us about the posture of a cow, whereas multiple frames can show how the cow moves over time. It is important since there are activities that might be quite similar in one frame only, for example, standing and walking, which can look almost the same in a single frame. So, the model should use a suitable amount of time for the detection task.

Identification and Re-ID are two distinct tasks. Closed set classifier identifies the cows seen in the training dataset. On the other hand, the query image is compared with the gallery images based on the embedding learned in Re-ID. It becomes possible for the system to be tested on cows unseen during the training phase. The good Re-ID embedding will classify images of the same cow into one cluster and images of the other cows into different clusters. The significance of this technique in cattle research lies in the changing herd of cattle [24].

2.1.3 Multi-Task Learning and Negative Transfer

Multi-task learning entails the training of different tasks in the same model where particular parts of the model are common and the training objective is minimizing the total loss of all the tasks,

$$\mathcal{L}_{\text{MTL}} = \sum_{t=1}^T \lambda_t \mathcal{L}_t, \quad (2.1)$$

where in this equation, $\mathcal{L}_t$ is the loss of task t , and λ_t determines the weight of the loss of task t . This equation calculates the total loss; however, but there is no information about what amount of the network should be shared. A neural network model can share all layers, a few layers, or exchange information between tasks.

In the survey paper by Crawshaw [20], MTL is considered in the context of information sharing, task-specific components, and optimization. MTL could be advantageous if one task is providing some useful information for another task. However, joint training could be harmful to a specific task; that is, the effect is referred to as negative transfer. It might occur due to the inappropriateness of information shared between tasks, conflicting task gradients, and unbalanced training losses. It is critical to distinguish information sharing and training for the purpose of the thesis.

2.2 Review of Existing Research

2.2.1 From Automated Observations to Integrated Livestock Monitoring

Livestock-monitoring research has moved from separate measurements toward systems that combine sensing, prediction, and decision support. This can be seen in both computer-vision reviews and broader precision-livestock studies. Antognoli et al. [38] review computer vision in dairy management, while Zin and Tin [51] discuss computer vision across wider livestock applications. These reviews show that image recognition is only one part of monitoring. The image first needs to be turned into a reliable observation about the animal before it can support a farm decision.

Research on livestock data analysis also shows why integration is important. Palma et al. [42] report that predictive analysis is much more common than prescriptive decision support in the dairy studies they review. Sani et al. [48] also discuss the difficulty of moving precision livestock technologies into practical use. This means that collecting more data or improving one classifier is not enough. The outputs also need to be connected over time and related to the correct animal and production setting.

Sensor-based and vision-based systems provide different information. Lee et al. [47] discuss how biosensors can provide continuous measurements at the individual-animal level, while cameras directly record visible appearance and activity. Vision is especially useful for this thesis because body shape, posture, movement, and coat patterns are all visible in images or video. This gives the three tasks a common sensing source, even though they need different information from it.

It is not necessary for a unified system for cattle monitoring to have exactly the same components for all the tasks. Some of the components of the system, such as cameras, preprocessing, and common features, can be shared among the tasks, while others remain specific to each individual task. Therefore, the integration is not only about merging multiple predictions. It also tries to find which features are useful for all tasks and which features should stay task-specific.

2.2.2 Body Condition Scoring: Anatomy, Image Geometry, and Ordered Prediction

BCS Automation is highly dependent on the cow's anatomy. Edmonson et al. [1] proposed an index of scoring while Ferguson et al. [2] investigated the correlation between anatomical traits and certain BCS values. From a point of view of computer vision, considering the whole body mass of the cow is not sufficient. In order for the model to work, it should detect the amount of tissue covering the body, body shape, and specific body parts. This is why the camera position and selected body area matter.

Another approach uses depth images for better representation of the body shape of a cow. In particular, Rodriguez Alvarez et al. [13] employ convolutional neural networks along with depth images and validate the prediction of BCS by means of various error thresholds. This approach relies on the combination of body geometry and features rather than the handcrafted features alone. The paper also proves that exact accuracy and tolerance-based accuracy have to be evaluated separately

because of the fact that BCS scores are ordinal.

Another methodology makes use of ordinary RGB images to detect visual cues associated with body condition. In the case of Liu et al. [41], most of the feature extraction takes place in the tail area and is done using the EfficientNet-B0 architecture with channel and spatial attention. Their work involves the selection of a specific body part and an efficient network architecture, while distillation assists in making the model size smaller. The fact that BCS prediction can be tailored through a combination of the specific body part and architecture has been illustrated.

Camera view influences the information that can be obtained by the model as well. Yao et al. [49] select side-view pictures of the cattle using a ConvNeXt-based regression model and cow-level cross-validation. One method is focused on the tail of the animal; therefore, it observes just a small portion of the body, whereas another method selects side-view pictures and considers a bigger part of the cow's body. Both methods are different not only in their models but also in the parts of the cow they observe.

The research done on cattle measurement also reveals that there is a need for distinction between image size and true physical measurements. The authors Guzhva et al. [46] came up with a system called PickAMoo, which uses smartphone images, LiDAR scaling, and Mask R-CNN segmentation in order to estimate weight. Even though the purpose of estimating the weight is different from estimating BCS, this method is relevant since the estimation is done when the animal has been separated from the image, and pixels are related to physical measurements.

The common factor among all these papers can be seen in three aspects that make good components for BCS prediction. First is the use of images that have proper features of the body and its orientation. Second is the use of an appropriate technique to extract such information, for example, cropping, body contour, depth map, or attention. Third is selecting an appropriate prediction technique that is suited to ordered BCS. CORAL can handle ordering of output classes [19], but it will not perform well if the input class does not have appropriate condition features. Thus, body-shape information will be considered as part of the representation, and ordinal prediction as the model choice.

2.2.3 Behavior Recognition: Posture, Motion, and Environmental Context

Behavior classes can differ in both appearance and duration. Lying may often be recognized from body posture, while walking is better described by movement. Feeding and drinking also involve interaction with a resource such as a trough. Because of this, behavior research uses different amounts of spatial and temporal information, and a single-frame classifier cannot describe every type of activity equally well.

Image-based behavior recognition shows that posture can already provide useful information. Asim et al. [44] use overhead-camera images and object-detection models to recognize activities including standing, lying, grazing, and estrus-related behavior. Their method is vision-based rather than wearable-sensor based. It shows one way to connect animal localization and activity labels in a frame. A sequence model can then add information about how those visible postures change over time.

Video datasets extend the task beyond one frame. CBVD-5, introduced by Li et al. [33], contains standing, lying, foraging, rumination, and drinking. CVB, intro-

duced by Zia et al. [32], provides densely sampled behavior clips from an open-pasture setting and includes a SlowFast benchmark. The datasets have different environments and behavior labels, but both show that behavior recognition depends on the animal and on how the activity is defined and observed over time.

MmCows adds synchronized sensors and individual-animal information to video data [35]. Its annotations include identity, location, and behavior across several camera views. This allows researchers to ask whether behavior predictions stay consistent across views and animals. A dense video dataset with only temporary tracking IDs can still provide useful motion examples, but it does not provide the same level of biological identity information.

These demonstrate two different aspects of behavioral data: temporal information and identity consistency. Video in high density helps with motion analysis in short term. Identity of the cows helps with the test being done on different cows and across time. The one does not substitute for the other. This means that a study that requires both needs to factor in both and not just choose the one with many images.

Contextual background information can also be useful and detrimental. While a dip or feeding barrier could aid in learning about feeding/drinking, the problem is that it might learn that one task tends to happen in one camera quadrant. While the removal of the entire background will make the shortcut less obvious, it will also take out some interaction information. It is not about whether the context can be seen as positive or negative. What matters is the amount of context that is helpful to comprehend the activity and its ability to remain relevant despite changes in the environment.

This is linked to the cow-focused thesis in which the cow is still the key element of the visual reference and where posture, movement and context relationship can all be used if it is necessary. It also allows for the use of a class-based assessment – a model can perform well on common rest behaviours and poorly on rare or more dynamic ones. This can be obscured by the overall accuracy if class frequency is correlated with source.

2.2.4 Individual Identification and Re-ID: From Coat Patterns to Cross-Setting Recognition

The visual identification of cattle requires certain characteristics that differ from one animal to another. Andrew et al. [6] use the technique of local coat pattern matching through RGB-D images. This technique provides us with an example of how the useful characteristic of identity is made explicit: local patterns are able to identify Holstein-Friesian cattle. This technique also indicates an important drawback that remains relevant for today's models. The identity-related feature must be visible in both images that are being compared.

Deep metric learning changes how these comparisons are learned. Andrew et al. [24] use an embedding-based method with a combined classification and reciprocal-triplet objective. Instead of only using a classifier for known cows, the embedding supports similarity-based matching. This allows the feature space to be tested on cows that were not used to train the encoder. The goal therefore changes from memorizing fixed identity labels to learning features that can separate animals more generally. Face-based identification uses a different body region. Weng et al. [28] use a two-

branch convolutional network to combine cattle-face information. Compared with coat-based methods, this focuses on facial differences between cows. Which approach is more useful depends on the camera view. A clear face may be hard to obtain from an overhead or rear view, while a useful coat pattern may be hidden by occlusion or missing from a tight face crop. Camera position and visible body coverage are therefore important parts of Re-ID.

Another line of work reduces the amount of manual identity labeling needed in video. Gao et al. [25] use the Cows2021 dataset to study self-supervision from tracking-based observations and normalized cow views. Yu et al. [43] extend this idea to several cameras on a working farm with MultiCamCows2024. These methods use the continuity of video to connect images that are likely to show the same cow. A short tracklet can therefore help the model learn appearance consistency without requiring a manually assigned cow ID for every frame.

Nevertheless, tracking and Re-ID are still different tasks. While tracking maintains association of an animal when it stays continuously visible, re-identification must restore this association when there is a gap in time or change in viewing conditions. Information about tracking may be used for training of the Re-ID model, but an id of the temporary tracking does not correspond to biological ID of the cow. Multi-camera and cross-setting assessment would be helpful since it examines bigger shifts than a continuous video sequence.

Another challenge is long-term identification. In the dataset called BECA proposed by Zhang et al. [50], there is a large population setting for beef cattle and a longitudinal setting. Unlike short-term pairing of dairy cattle with unique coat patterns, an animal may look different depending on its size, its stance, and the part of the body that is visible, and having a larger population increases visual similarity. Thus, the Re-ID model needs to be tested not just on the cows used in the training phase, but also on its ability to identify across populations and time.

Similarly, foreground processing and anatomy processing have to take into account the need to preserve the data needed for identification purposes. While deleting the background may force the model to concentrate on the cow, deleting too much of it by reducing it to just a silhouette or skeleton may lead to loss of critical coat features. Retaining the full environment will cause the opposite effect by teaching the model the usual position of the cow. Consequently, Re-ID will require a selective representation technique that minimizes environmental shortcuts but maintains cow-specific characteristics.

2.2.5 Localization and Segmentation as Representation Design

The localization determines the cow on which the model needs to be applied. This becomes very crucial when there are multiple cows in the same image since the predicted class or behavior belongs to that particular cow only. The bounding box provides an effective ROI, but also maintains some context within the image. Image cropping can normalize the scales and enable the encoder to focus, but alters the body-background ratio.

There are several modern object detectors that can help to locate this region. RT-DETR developed by Zhao et al. [37] uses an end-to-end transformer detector optimized for speed-accuracy trade-off. Instance segmentation approaches are richer

than bounding box. Mask R-CNN extends the detection network with the mask branch and thus makes it predict not only the position but also object specific mask [7]. To be more precise, the bounding box says where the cow is, while the mask defines its pixels.

Promptable segmentation decouples the process of selecting the target from creating its mask. Promptable segmentation model, Segment Anything by Kirillov et al. [30], is capable of generating masks for prompts such as points and boxes. Promptable segmentation in SAM 2, being memory-based, is applicable to both images and video [34]. This implies that in a cattle pipeline, for instance, we can first detect the cow, and then prompt the segmentation model to generate the mask, without the need to train a new cattle-specific segmentation model from scratch. The final mask will depend upon how the subject was chosen and prompted.

Cattle-specific datasets provide more direct supervision. CattleEyeView includes top-down annotations for detection, counting, tracking, pose, and instance segmentation [31]. Because several types of annotation describe the same scenes, it is useful for studying cattle perception. However, these perception tasks are not the same as predicting BCS, behavior, and identity. For this thesis, they are useful ways of describing the cow, not proof that one description should be shared by every downstream task.

Recent cattle recognition work also links foreground estimation with identity learning. CATR-Net combines segmentation and cattle-face recognition using edge-based segmentation and attention-based recognition components [40]. This directly connects target separation with identity features. Together with earlier coat- and face-based methods, it shows that cattle-centered recognition is already an active research area. The open question for a broader monitoring system is whether foreground emphasis also helps tasks that depend more on morphology or motion.

Segmentation and feature attention are also different. CBAM changes the importance of channel and spatial features inside a network [14], while a foreground mask changes or weights the image content using an estimated object boundary. Attention can focus on useful regions while still keeping the full image. Masking can reduce background information before the main encoder. The two methods may work together, but they are not the same intervention.

Masking could also be applied with different degrees of strength. With hard foreground masking, all content outside of the estimated boundary is eliminated. The softer form of masking will lessen the effect of the background while retaining some context along with uncertain boundary information. It is a matter of selecting the appropriate approach for the particular task. Useful body contours are required for BCS, context is needed for behavior, and coat information for Re-ID. Therefore, the quality of a mask should be assessed together with the performance of the downstream task, rather than only by how clean the mask appears.

2.2.6 Pose, Anatomy, and Viewpoint

Pose estimation represents an animal using the positions of selected body parts. DeepLabCut shows that user-defined landmarks can be learned with transfer learning from a relatively small amount of annotation [12]. This changes the input from dense appearance information to a structured description of body configuration. Such information can support movement analysis and can also make some parts of

the model input easier to understand.

SuperAnimal extends this idea by combining different animal-pose datasets into transferable models [36]. Its training method handles differences between landmark definitions and supports both zero-shot use and further adaptation. This is useful for cattle work because it reduces the need to create a completely new pose dataset from the beginning. CattleEyeView provides a more cattle-specific option in a top-down setting [31].

These works taken together indicate that there are two ways in which structural information can be acquired, either through transfer learning from many different animal datasets or through training on a cattle dataset. The definition of landmarks is just as important as the pose estimation model. The skeletal information can be obtained through the head, trunk, and legs, but BCS involves more than just pose estimation. Thus, the BCS descriptors analyzed by Ferguson et al. [2] must necessarily include not only joint positions. The pose and the structure are related but are certainly not the same representation. BCS might require both structure coordinates as well as visual features, whereas the behavior may rely more on coordinate changes.

Viewpoint decides which body parts are visible and how the cow appears in the image. Rear, side, and overhead views can show different body contours even when the cow's real condition has not changed. This matters for Re-ID because the cow should still be matched across views, and for BCS because important regions may be hidden. Viewpoint means the orientation of the animal relative to the camera; it is not the same as camera identity.

The MOO preprint by Grolleau et al. [45] studies viewpoint using synthetic cattle rendered from controlled directions. Its angle labels make it possible to study geometric changes separately from some uncontrolled changes in natural images. The authors also study transfer to real cattle images. Synthetic data therefore provide controlled viewpoint labels, while real images test whether the learned information still works under natural appearance and environmental changes.

The main idea is that invariance should be selective. A Re-ID representation may need to stay stable when the same cow is seen from another angle, but a behavior representation should still react when the posture changes. BCS may need stable body-shape information even when image scale changes. Pose and viewpoint can help describe these changes, but the useful response depends on the task. This is why the thesis studies task-conditioned structural information instead of forcing one invariant representation on every output.

2.2.7 Temporal Representations for Activity Recognition

Temporal modeling provides the information related to the changes through time. This information varies depending on representation choice. RGB sequences store information about the visual aspect and posture of the subject, while keypoints sequences provide a compact representation of the body configuration. Video modeling allows for learning both spatial and temporal representations at the same time. These approaches preserve different information and require different computational and labeling effort.

SlowFast network explicitly separates the aspects of appearance and motion using two streams with different frame rates [16], where the first stream concentrates on

Table 2.1: Visual information and task requirements synthesized from the reviewed literature. The entries describe design considerations, not measured outcomes of the proposed system.

Visual cue	BCS	Behavior	Re-ID
Foreground appearance	Body contours and local condition cues	Posture and interaction details	Coat markings and individual morphology
Pose and anatomy	Region alignment; joint locations alone may omit condition cues	Body configuration and its changes	Possible alignment aid; limited identity texture
Viewpoint	Visibility of condition-related regions	Interpretation of projected posture	Correspondence across views
Temporal context	Repeated observations of condition	Motion, duration, and transitions	Appearance continuity across observations
Scene context	Often secondary to body evidence	Potentially useful near feed or water resources	Possible background-dependent matching

spatial information and the second captures faster changes. SlowFast network is also used as the benchmarking technique in the CVB application [32], making connection between general video recognition technique and behavior of cattle. It provides a stronger motivation for temporal modeling approach other than simple assumption of more being better.

A temporal convolution network may also be used. In comparing convolutional and recurrent sequence models, Bai et al. [8] found that temporal convolutions have shown promise in several sequence tasks. These models leverage temporal properties without needing to include a recurrent hidden state at each step. In relation to cattle behavior, this provides justification for using a light temporal network in place of image features when many visualizations must be compared.

Another way to encode motion is to consider body landmark relations over time. ST-GCN by Yan et al. [15] is a model that uses graphs to represent the skeleton of a moving human being with connections between nodes in space and in time. This approach is applicable to the problem of interest due to modeling relations between body parts rather than modeling coordinates separately. Still, its use in the case of cattle depends on having correct landmarks definition and accurate estimation of poses. Therefore, the pose sequence should be seen as a different representation category and not a substitution for RGB video.

The approaches prove that different temporal inputs address different problems. Taking the average of several frame features provides the model with more observable appearance, while a temporal model can also make use of the sequence of those frames. Pose features concentrate on body pose, while dense video contains more motion and contextual information. This choice is based on the type of action and sampling period. Hence, this thesis makes use of temporal modeling as a representation problem rather than scaling up the model.

2.2.8 Shortcut Learning, Domain Shift, and Evaluation

Success of the algorithm in a certain known situation does not mean that the representation learned by it will be useful in another place. Geirhos et al. [21] demonstrate that deep learning models employ some simple correlations that cannot be applied when the environment is altered. It becomes especially crucial if the object and background are seen in many cases. For instance, in cattle pictures, the same pen, lighting, and viewpoint can occur repeatedly, this makes it easier for a model to find scene-specific shortcuts.

Background dependence can be tested directly. Xiao et al. [27] change foreground and background information to study their effect on image recognition. Their work shows why evaluation should go beyond testing on an unchanged dataset. For cattle-centered learning, the question is whether the model still depends on the cow when irrelevant parts of the scene change. Foreground processing is one way to test this, while changes in camera, setting, or dataset provide other robustness tests.

Animal-recognition research shows that this problem also appears in real-world settings. Beery et al. [9] report a large performance gap between familiar and unfamiliar camera-trap locations. Wildlife camera traps are different from livestock cameras, but both can contain repeated fixed backgrounds. WILDS extends this idea with benchmarks that include natural distribution shifts across wildlife and other applications [26]. These studies support evaluating the conditions under which a model should generalize instead of treating every held-out image as equally difficult.

Domain shift can involve more than the background. It may include new cows, different breeds, changes in body appearance, new camera heights, or even different label definitions. Re-ID studies test some of these changes through multi-camera and longitudinal settings [43], [50]. BCS studies may use rear, side, or depth views, while behavior datasets may use different activity definitions and environments. External evaluation therefore needs to consider both the image conditions and whether the labels mean the same thing.

Evaluation samples also need to be properly separated. Kapoor and Narayanan [29] show that data leakage can make machine-learning results look better than they really are. In video data, two different image files may still come from the same continuous event. A random image split can therefore place nearly identical situations on both sides of the train-test boundary. Grouping by the correct unit is safer. The correct unit depends on the claim: unseen-cow evaluation needs reliable cow IDs, while recording-based grouping protects against a different type of overlap.

In this sense, the design of the evaluation procedure is part of the representation problem. The mask will seem relevant in one split since the model is dealing with an easy identity and localization task. The use of multiple datasets will bring variation visually while one dataset is still linked to one class. Evaluation and training should be aligned with what is claimed in this case. This brings more validity to testing whether the use of cattle information helps with transfer rather than improving performance within one known data set.

2.2.9 Multi-Task Learning: Selective Sharing and Negative Transfer

The use of multitask learning (MTL) allows the combining of the three models into one system. The most basic design uses a single encoder and different heads for

each task. This helps in saving computational resources and encourages the use of common features. However, the use of MTL requires more than the fact that there are three inputs containing cattle. Crawshaw [20] notes that architecture, task relationships, and optimization all have roles in MTL performance. Thus, the sharing of the resources should have something more compelling than just the fact that the three inputs have cattle.

Some of the methods used in MTL attempt to ensure the balance of task updates. In Kendall et al. [11], loss weighting is done using task uncertainty. The other technique is called GradNorm, which uses gradients for modifying the task weight depending on learning pace [10]. These techniques seek to prevent dominance of a particular task during joint learning due to either loss magnitude or pace. They control the strength of each task's update to the network. However, they do not select features to be shared visually.

PCGrad solves a completely different aspect of the problem through the modification of gradients during conflicting task updates [23]. Thus, gradient interaction becomes a component of the optimization process. Modification of the magnitude of an update and modification of a conflicting direction are two separate things, and therefore, PCGrad cannot be equated to loss weighting. Both these techniques remain within a selected architecture, thus allowing researchers to explore optimization as one of the causes of negative transfer.

Other approaches modify how the features are shared. Cross-Stitch Networks teach how to share the activation of task-specific networks [5]. The Multi-Task Attention Network proposed by Liu et al. [17] uses task-specific attention to choose information from a common pool of features. Such approaches do not enforce that all tasks use the same feature route. This is especially helpful when tasks are similar but require different information, such as for cattle morphology, behavior, and individuality.

Task grouping can also be tested instead of decided in advance. Standley et al. [22] study which tasks should be trained together under limited resources. Their work shows that task relationships cannot be decided only from task names or descriptions. Two tasks may share early visual features but need separate processing later, or one grouping may help one task while hurting another. Strong single-task references are therefore necessary before judging a shared model.

Animal-focused work shows both the value and the limits of joint learning. SAAM-VetNet combines disease detection and severity grading using an attention-based MTL framework [39]. CATR-Net combines cattle-face segmentation and recognition [40]. These studies show that MTL and perception-guided recognition are already useful in animal vision. However, their task relationships are different from those in this thesis. Segmentation and face recognition use the same visible target, while disease type and severity may use similar local signs. BCS, behavior, and Re-ID need more different kinds of information.

For example, Re-ID needs stable differences between cows. Behavior recognition must still respond when the same cow changes activity. BCS needs morphology but should not replace condition information with cow-specific appearance. A shared encoder can support all three only if it keeps these differences. This is why the thesis compares hard sharing with task-conditioned or partly private processing.

Using separate datasets for the tasks creates another issue. Joint training sees differences in camera view, image composition, and annotation style at the same time as it sees different task objectives. Some apparent task differences may therefore

Table 2.2: Complementary approaches to managing multi-task interaction. The methods address different aspects of joint learning and therefore support different interpretations of an improvement.

Approach	What changes	Question addressed
Uncertainty weighting [11]	Relative contributions of task losses	How should losses with different uncertainty be balanced?
GradNorm [10]	Adaptive task weights using gradients	How should task learning rates be balanced?
PCGrad [23]	Interaction of conflicting updates	Can harmful gradient interference be reduced?
Cross-stitch sharing [5]	Exchange between task-specific features	Which feature combinations should be shared?
MTAN [17]	Task-dependent selection of common features	Which shared information is useful to each output?

actually come from the datasets. A careful comparison must keep each task protocol fixed and examine every output separately. This makes it possible to test whether selective sharing helps the integrated model rather than simply giving it more parameters.

2.3 Summary of Key Findings

The literature shows that the research problem is more specific than simply combining three cattle classifiers. BCS, behavior, and Re-ID need different visual information. Anatomy-based scoring, temporal behavior models, and metric-learning Re-ID each solve part of the problem, but they focus on different properties of the cow. A unified framework therefore needs to keep these differences in mind.

2.3.1 A Common Animal, but Different Information Requirements

BCS research mainly focuses on anatomical coverage, body geometry, and the ordered BCS target. Behavior research adds movement, duration, and interaction context. Re-ID research focuses on keeping individual appearance consistent across different observations. These goals can support each other, but they can also conflict. Removing individual texture may help BCS avoid identity shortcuts, while Re-ID needs that texture. Scene context may help recognize feeding, while the same context can create background shortcuts for identity.

Previous research provides many ways to localize cattle, describe their body structure, and combine perception with recognition. The research gap is understanding how these choices affect BCS, behavior, and Re-ID differently, and how task-appropriate cattle-centered representations compare with generic RGB features when the tasks are brought into one shared system.

2.3.2 Representation Quality Must Be Connected to Robust Evaluation

Perception models can provide masks, landmarks, and viewpoint information, but producing an output is not enough. The representation must still keep the information needed by the task when relevant conditions change. For BCS, this includes visible anatomy and acquisition conditions. For behavior, it includes posture, motion, and scene changes. For Re-ID, it includes different views of the same cow across place or time.

This creates a second research gap: representation changes need to be tested under evaluation conditions that match the expected type of change. Foreground-focused models should be judged together with cross-setting and class-sensitive results, not only with one pooled score. Temporal models should also be evaluated in relation to how the observations are sampled over time. The thesis is therefore organized around the connection between visual information, task needs, and generalization conditions.

2.3.3 Selective Sharing as the Basis of the Unified Framework

MTL research separates two questions: how task updates are balanced, and what information is shared between tasks. The preliminary investigations in this thesis make this distinction important for cattle monitoring. Instead of assuming that heavy sharing is best, the study first uses single-task references to understand the representation needs and then compares different ways of combining the tasks.

The research approach can be seen as a task-specific cattle-oriented approach. It seeks to preserve valuable common attributes of the cattle while being flexible enough to place different levels of attention on morphology, behavior, and identification. The comparison is mostly going to be made between generic RGB baselines and hard sharing approaches. In particular, the task performance and negative transfer will be of the main interest. The study also aims to identify which features can be shared, which ones require task-specific processing, and how these decisions influence the unified system.

The literature review above serves as the foundation for subsequent chapters. Chapter 3 outlines the criteria and constraints of the task. Chapter 4 explains the protocols used for the data, their representations, and model comparisons. Chapter 5 applies the results gathered to readdress the research questions.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

The framework is meant to give three results: an estimate of the animal's body condition, a label for its behavior, and a representation of its identity for retrieval. As a research prototype, not as a finished farm product, it is being looked at. So, the requirements include both functions that make predictions and the controls that make sure the results are properly interpreted. Some of these are clear label definitions, protected data splits, failure handling, task-specific metrics, and settings that allow the results to be reproduced.

Table 3.1: Research requirements and the evidence needed for acceptance.

ID	Requirement	Evidence required
R1	BCS output with explicit units	Physical-score mapping, output support and saved metrics.
R2	Target-cow Behavior output	Taxonomy, crop/cache integrity and class-wise results.
R3	Re-ID under a named protocol	Identity partition, query/gallery definition and retrieval artifact.
R4	Evaluated integrated model	Matched single-task / E1 / E3 / E4 comparisons and documented sharing / optimization boundaries.
R5	Provenance-aware evaluation	Grouped splits, versioned checks and limitations.
R6	Reproducible execution	Executed settings, seed, revision, hashes and checkpoint.
R7	Transparent failures	Missing-input/perception counts; no silent substitutions.
R8	Feasible implementation	Measured resources and maintainable dependencies.

For BCS, the model must predict within the documented score range. The behavior model must use the fixed five-class label set. The Re-ID model must clearly define the learning identities, gallery, queries, and retrieval metrics. Requirement R4 covers

the final integration of all three tasks into one multi-task model, including a clear description of what is shared and what remains task-specific.

The main non-functional requirements are reproducibility, leakage resistance, maintainability, clear failure reporting, and practical execution. Dataset manifests, scripts, metrics, and configurations are versioned. Raw datasets and large checkpoints are stored separately. Reproducing an experiment therefore requires both the recorded configuration and access to the correct data.

Scientific validity adds further requirements. Test data must not be used for checkpoint selection. If perception failures remove samples, comparisons must either use matched samples or report the coverage clearly. Missing task labels must be handled with task-specific batches or another available-label method, not treated as negative labels. Finally, every evaluation claim must match the unit protected by the split, such as a burst group, source video, recording session, or biological identity.

3.2 Societal Impact

A common visual monitoring system may be useful to organize repeated observation of cattle by associating body condition and behavior to individual animals. This can help people see unusual changes earlier, and reduce the amount of manual video inspection that has to be done in practice. However, this study does not incorporate farmer interviews, field deployment or evaluation of labor savings, productivity, treatment outcomes or benefits in animal welfare. Those are prospective benefits, not measured results.

The model should assist human decision making, not replace it. Poor identification matching can lead to observations being assigned to the wrong cow, and an inaccurate BCS or behavior forecast can cause to a delay in attention. A realistic system has to save the original image, the time, the confidence or proof of retrieval and a way for a person to judge the output. There still has to be good professional judgment in veterinary and welfare decisions.

Conditions at the farm will be an important aspect when using the system. Aspects like camera placement, accommodation, breeds, lighting, internet, and staff working procedure may vary from farm to farm. Due to this, the current findings cannot confirm the system's success in all areas including Bangladesh. Tests must be done by people in relative fields before claiming that the system works in real world.

Agricultural images may depict people, arrangements, or personal information. There is no guarantee in the license to utilize a database in research that the images may be distributed or used commercially. Privacy, ownership, and licensing regulations must be considered for each database and each potential field use.

3.3 Environmental Impact

The cost of perception pre-processing, model training, data storage, and data transmission is included in the computational cost of the system. It is not confined to the prediction model. An increased detector or segmentation model may have an impact on a smaller model at a later stage. This should be made explicit due to its possible effect on the efficiency of the system.

We reduced repeated computation by checking the pipeline before full experiments, reusing validated cached outputs, and using deterministic manifests. These steps reduce unnecessary repeated processing, but we did not collect a complete power trace or location-specific emissions measurement. Therefore, the study does not claim measured carbon savings or measured emissions.

Better monitoring could eventually support more efficient livestock management. However, the experiments do not measure feed savings, methane reduction, or changes in disease burden. Those effects would depend on how the predictions change real farm decisions and would need a separate long-term study.

3.4 Ethical Issues

A major ethical responsibility in this work is to separate what the data actually show from what we might assume. ScienceDB is not described as cow-disjoint because verified biological cow IDs are not available. The primary behavior evaluation is not described as unseen-cow evaluation. The oracle segmentation-guided Re-ID experiment uses ground-truth masks, so it is not evidence of an automatic segmentation system. Positive, mixed, and negative findings are all reported.

Dataset reuse must also be checked for each source. The dataset registry records the available licenses and access conditions, but any unresolved permission must be confirmed before redistribution or deployment. The thesis does not claim institutional ethics approval, an exemption, a user study, or field consent unless a supporting institutional record exists.

The model selection is done using validation data, while the evaluation is done using held-out test data. Do not use external test data for tuning after you have seen the results. Coverage information is used to report samples that are discarded due to perception failures, rather than being silently ignored.

The authors have to be responsible for the final product even if the use AI in writing and programming the project. The authors need to ensure that the project is accurate, all facts are checked, AI policies of their institution are adhered to, and the correct citations are given.

3.5 Standards and Technical Conventions

The study follows common technical practices. However, no claim is made about having an engineering certification unless it has been properly verified. The released score range and the known anatomical meaning of BCS are used to understand what the BCS labels represent [1], [2]. Labels for behavior use a versioned mapping, and Re-ID has a clear query-gallery protocol. That being said, these rules do make the evaluation more fair, but they do not officially certify the method.

The thesis is prepared according to the prescribed chapter format of BRAC University and bibliography style of IEEE. Dataset licensing, privacy requirements, deployment safety and any separate IEEE-format submission requirement need to be reviewed independently. The use of IEEE reference format does not imply that the work has been submitted or approved for publication to the Journal.

3.6 Project Management Plan

The project is organized into several main stages: protocol verification, perception assessment, single-task reference models, cattle-centered alternatives, and multi-task comparison. Each stage provides the information needed for the next. Table 3.2 records the main project milestones. The timeline shows project progress, not scientific evidence.

Table 3.2: Research milestones and manuscript-integration stages. Recorded dates are retained from the existing evidence summary.^[E01,E02,E03,E23,E25,E28,E30]

Stage	Milestone	Record or deliverable
Preliminary work	Problem formulation and initial investigations	Research motivation
20 Sep 2026	ScienceDB grouping and SideView protocols	Recorded protocols
20–22 Sep	Perception and visual assessments	Feasibility findings
23 Sep	CVB and Beef protocol and RGB results	Recorded comparisons
23 Sep	RGB baselines and SideView data preparation	Completed single-task references
24 Sep	Cattle-centered task models	Completed task evaluations
MTL comparison	Hard-shared control, modular task-private model, and PCGrad optimization control	Completed held-out evaluations
Final synthesis	Integrated discussion, abstract, and conclusion	Completed manuscript synthesis

Computational resources were selected according to the needs of preprocessing, training, and evaluation. Small pipeline checks were done before full experiments, and reusable intermediate outputs reduced repeated preprocessing. Detailed hardware and execution records are kept for reproducibility instead of being included as part of the main scientific argument.

With versioned code, deterministic manifests, permanent data storage, and machine-readable results, each experiment is linked to its own set of settings. These records help make sure that the same steps are taken in all situations and give us the data we need to repeat the results that were published.

Contributions from authors, the order of authors, information about the committee, and submission details still need to be confirmed by the authors and supervisors. They are not guessed from what is happening in the repository.

3.7 Risk Management

There is a list of the main scientific and operational risks in Table 3.3. Some of these are related observations that cross train and test splits, identity claims that can't be backed up, classes that are linked to the same data source, perception failures, wrong causal claims from combined model changes, and unfair comparisons caused by different training budgets.

Table 3.3: Scientific and operational risks; implemented controls do not eliminate all residual risks.^[E04,E05,E06,E07,E11,E24,E26,E29]

Risk	Consequence	Control / open issue
Correlated data	Optimistic evaluation	Burst/source/session grouping; identity may remain unknown.
Temporary IDs	False cross-cow claim	Use biological IDs only where verified.
Unavailable data	Blocked experiments	Approved SideView contingency; MultiCam still blocked.
Source/class link	Shortcut prediction	Source/class metrics; Walking is CVB-only.
Perception error	Missing/wrong target input	Track valid masks and omit failed samples without fabricating data.
Combined changes	Unsupported causal claim	Report performance for the whole setup rather than single parts.
Oracle masks	Overstated deployment readiness	Mark Run 6 as using ground-truth masks, not automatic tools.
Single-run results	Unknown training variability	Give single-run numbers without claiming statistical significance.
External mismatch	Misleading transfer	Lock label groupings prior to evaluation.
Limited resources	Incomplete comparisons	Run quick local checks, train in cloud, and postpone E2, E5, and stress tests.
Unclear license	Uncertain reuse rights	Check data licenses directly instead of assuming rights from web access.

Task-specific controls in the evaluation design lower these risks. ScienceDB burst groups and behavior videos or sessions stay in the partitions that were given to them. SideView Protocol A separates biological identities between representation learning and evaluation. We use matched subsets to check how the RGB, perception-enhanced BCS, and behavior models work on the same samples that were kept. When ground-truth masks are used, the Re-ID model is clearly called "oracle-guided." There are still some important limits: Individual cow identities in ScienceDB are unknown, Walking is linked to a single source, and the task models are based on a single run. The final MTL comparison needs to keep the task protocols comparable and report any changes in the training budget.

3.8 Economic Analysis

The available records allow only a partial discussion of research cost, not a full farm-deployment business case. The experiments use different accelerators, preprocessing steps, storage needs, and billing conditions. Their runtime therefore cannot be turned into directly comparable cost values without consistent rates or invoices. Table 3.4 reports only information that was actually recorded.

Table 3.4: Selected recorded training/pipeline runtimes; these are not a total project or deployment budget.^[E08,E09,E14,E22,E24,E28,E29]

Item	Elapsed time	Cost evidence
BCS baseline	2,023.22 s	About USD 1.15 in research log; not invoice-reconciled.
Behavior baseline	821.47 s	Dollar costs were not tracked in the result file.
Re-ID RGB baseline	1,049.05 s training	No financial cost was saved in the artifact log.
BCS cattle-centered	550.80 s across recorded epochs	Vision preprocessing was run separately without an overall total.
Behavior cattle-centered	1,009.91 s training	Offline data caching took 3,492.3 seconds as a separate run.
Re-ID oracle GT-guided	2,350.67 s full pipeline	Covers the training time and Protocol A test evaluation.
Complete system	Not measured	Commercial payback and power usage were not measured.

To measure the cost of deploying the model for the practical purpose, it is important to understand how many cameras are used, their location, frequency of the inference process, required hardware, data storage, and maintenance. The present benchmark tests do not provide this data. As a result, there are no reliable cost savings, returns on investment, or payback periods.

Chapter 4

Proposed Methodology

4.1 Design Process and Methodology Overview

The study tests different ways of showing the visual representation to the model and each task uses its own evaluation model. Each task has an RGB reference model and a cattle-centered version. These six single-task models establish task-specific performance baselines and serve as the empirical reference for evaluating three multi-task learning configurations: a monolithic hard-shared control architecture (E1), a modular task-private architecture (E3), and a gradient-projected hard-shared optimization control (E4).

The main design question is what information each task should receive. BCS mainly needs morphology, behavior recognition needs posture and temporal change, and Re-ID needs stable individual appearance. Because of this, the cattle-centered models all focus on the target cow, but they do not use exactly the same preprocessing. Figure 4.1 shows the overall comparison, and Table 4.1 lists the model configurations. Supporting records are indexed in Appendix A.

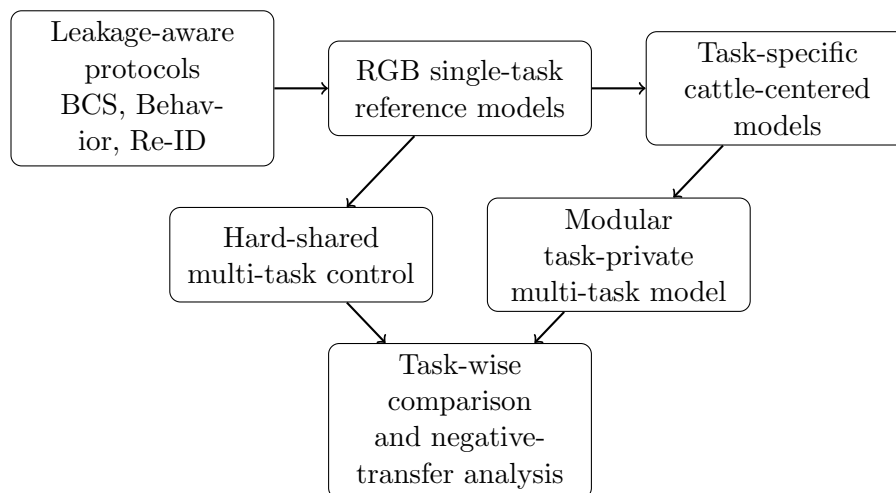


Figure 4.1: Comparative research framework illustrating single-task baselines, cattle-centered perception models, and unified multi-task evaluations across BCS, behavior recognition, and re-identification.

The perception-enhanced models change several things at the same time. The BCS model changes localization, cropping, and mask guidance. The behavior model

Table 4.1: Configurations in the comparative research framework. Single-task reference models and unified multi-task architectures are evaluated across the matched held-out protocols for body condition scoring, behavior recognition, and individual re-identification.

Configuration	Role	Evaluation Protocol
RGB BCS baseline	Single-task reference	Full 8,040-image test.
Single-frame RGB behavior baseline	Single-task reference	Full 809-sequence test.
RGB Re-ID baseline	Single-task reference	Protocol A Barn and Snapshot retrieval.
Perception-enhanced BCS	Crop and binary-mask guidance	Matched 7,549-image evaluation.
Perception-enhanced temporal behavior	Crop, binary mask, and Conv1D aggregation	Matched 780-sequence evaluation.
Oracle segmentation-guided Re-ID	Ground-truth crop and mask guidance	Protocol A Barn and Snapshot retrieval.
Hard-shared multi-task control (E1)	Monolithic joint representation reference	Matched multi-task evaluation across all three tasks.
Modular task-private multi-task model (E3)	Task-private residual adapter architecture	Matched multi-task evaluation across all three tasks.
PCGrad hard-shared optimization control (E4)	Gradient-projected shared representation control	Matched multi-task evaluation across all three tasks.

changes both the spatial representation and temporal aggregation. The Re-ID model changes crop geometry and adds an oracle mask channel. Therefore, these comparisons measure the full model configurations. They do not prove the isolated effect of any one component.

4.2 Preliminary Designs and Model Specifications

4.2.1 Representation Alternatives

The RGB reference models use ImageNet-pretrained ResNet-18 encoders so that the three tasks start from the same practical backbone [4]. The prediction head is different for each task: cumulative ordinal logits for BCS, categorical logits for behavior, and an identity classifier with a retrieval embedding for Re-ID. Keeping the task protocol fixed allows us to compare the RGB and cattle-centered inputs more fairly.

Localization and segmentation were chosen after testing pretrained perception models. The study checked detector return rate and latency, and measured mask overlap where SideView ground-truth masks were available. RT-DETR-L followed by SAM 2.1 gave a practical automatic pipeline for ScienceDB and behavior data. SideView's released masks allowed a separate oracle-guided Re-ID experiment [34], [37]. Figure 4.2 illustrates the resulting task-specific perception and input preparation pipelines across the three monitoring domains. Pose and viewpoint were not included in these first paired models because their task usefulness and transfer reliability had not been established at the time. This does not mean that they are rejected as separate later ablations.

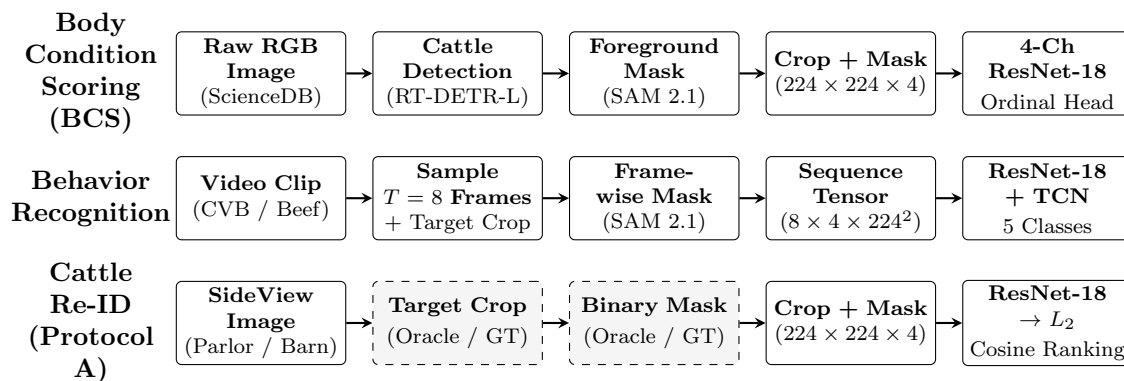


Figure 4.2: Task-specific cattle-centered input pipelines. Target cow localization and soft segmentation masks are generated automatically (RT-DETR-L and SAM 2.1) for BCS and behavior, and obtained from ground-truth annotations for re-identification.

4.2.2 Baseline RGB Body Condition Scoring

The BCS baseline uses an ImageNet-initialized ResNet-18 with four independent cumulative threshold logits for five ordered classes. For a class index $y \in \{0, 1, 2, 3, 4\}$,

the target for threshold k is

$$t_k = \mathbb{I}(y > k), \quad k = 0, 1, 2, 3. \quad (4.1)$$

If $g_k = \mathbf{w}_k^\top \mathbf{h} + b_k$ is the corresponding logit, the batch loss is

$$L_{\text{BCS}} = -\frac{1}{4B} \sum_{i=1}^B \sum_{k=0}^3 [t_{ik} \log \sigma(g_{ik}) + (1 - t_{ik}) \log(1 - \sigma(g_{ik}))]. \quad (4.2)$$

The predicted class is found by counting how many threshold probabilities are above 0.5. It is then mapped back to the physical BCS scale:

$$\hat{y} = \sum_{k=0}^3 \mathbb{I}(\sigma(g_k) > 0.5), \quad \widehat{\text{BCS}} = 3.25 + 0.25\hat{y}. \quad (4.3)$$

The targets are cumulative, but the model uses independent ordinal binary cross-entropy. Therefore, the method is described as `ordinal_bce`, not CORAL, because CORAL uses a different rank-consistency method [19]. The model was trained for 30 epochs with batch size 64, AdamW, an initial learning rate of 10^{-4} , weight decay 10^{-4} , a cosine learning-rate schedule, and seed 42. The checkpoint with the lowest validation MAE was kept.

4.2.3 Perception-Enhanced Body Condition Scoring

The perception-enhanced BCS model first localizes and segments the cow in ScienceDB. RT-DETR-L detects cattle using the COCO cattle class with a confidence threshold of 0.25. If more than one cow is detected, the largest bounding box is selected, and confidence is used to break a tie. The selected box is expanded by 5% in each direction, clipped to the image boundary, and used for both the RGB crop and the SAM 2.1 prompt. If detection or segmentation fails, the image is recorded as a perception failure and excluded. No artificial replacement input is used.

The RGB crop and binary mask are resized together to 224×224 . The RGB channels use ImageNet normalization, while the mask stays as an unnormalized binary channel in $\{0, 1\}$. The first ResNet-18 convolution is changed from three input channels to four. The pretrained RGB weights are kept, and the new mask-channel weights are initialized using the mean of the RGB filters. The ordinal head, target encoding, loss, training schedule, and augmentation policy otherwise follow the RGB BCS baseline. This adds 3,136 trainable parameters, increasing the model from 11,178,564 to 11,181,700 parameters. The model uses a binary mask channel; it is not a calibrated soft-mask model.

4.2.4 Baseline Single-Frame RGB Behavior Recognition

The behavior baseline uses the midpoint frame of each canonical sample. It uses an ImageNet-pretrained ResNet-18 with a five-class linear classifier. The five classes are Standing, Lying, Feeding, Drinking, and Walking, and the model is trained with cross-entropy loss. CVB samples use the target-tracklet information provided by the dataset, while Kaggle Beef clips already contain one main animal. The selected checkpoint is epoch 25 from the 30-epoch training run, chosen using validation Macro-F1.

This baseline intentionally does not use temporal aggregation. Its purpose is to show how much information one RGB frame can provide under the common five-class label set. For this reason, evaluation includes balanced accuracy and class-specific results as well as overall accuracy. This is especially important because Walking appears only in CVB.

4.2.5 Perception-Enhanced Temporal Behavior Recognition

The temporal behavior model uses eight approximately evenly spaced frames from each labeled interval instead of only the midpoint frame. Its input tensor has shape $[B, 8, 4, 224, 224]$. Each time step contains an RGB crop and a binary mask. For CVB, the target-tracklet box at the sampled frame is used to prompt SAM 2.1, and the same box is used for the RGB and mask crops. The model does not replace the target with a nearby cow or with the full image. For the single-cow Beef clips, RT-DETR-L provides the main cow box when available, followed by a positive center prompt to SAM 2.1. If the detector finds no cow, an image-center prompt is used. A sequence is excluded if any required mask is empty or invalid.

The training and validation cache kept 4,271 of 4,465 candidate sequences: 3,641 for training and 630 for validation. It contains 34,168 RGB frames and 34,168 non-trivial binary masks. There were 194 exclusions because of empty-mask failures, and all 145 Walking candidates in training and validation were kept. The held-out test comparison separately kept 780 of 809 samples, including 422 CVB and 358 Beef sequences.

Each frame is passed through a four-channel ResNet-18 to produce a 512-dimensional feature. The sequence $\mathbf{H} \in \mathbb{R}^{B \times 8 \times 512}$ is then transposed for temporal convolution. The first temporal block uses a width-three Conv1D projection from 512 to 256 channels, followed by BatchNorm, GELU, dropout 0.2, and a width-one residual projection. The second width-three block keeps 256 channels and uses the same normalization, activation, and dropout with an identity residual. Adaptive average pooling across time gives one 256-dimensional feature vector, which is passed to the five-class linear head. This is a lightweight residual Conv1D model over sparsely sampled frame features. It is not optical flow, a video transformer, or a deep dilated causal TCN. The full model has 11,903,621 trainable parameters.

4.2.6 Baseline RGB Cattle Re-Identification

The RGB Re-ID baseline follows SideView Protocol A. Representation learning uses 41 cows from the parlor setting only, with 12,753 training images and 2,683 validation images. Evaluation uses 69 different cows. The gallery contains 36,811 parlor images, the barn query set contains 25,260 images from all 69 cows, and the snapshot query set contains 607 images from 63 cows. The cows used for representation learning and evaluation do not overlap.

ResNet-18 produces a raw 512-dimensional feature $\mathbf{h}$. A 41-class linear classifier is trained on this feature with cross-entropy loss. Retrieval uses the L2-normalized version:

$$\mathbf{z} = \frac{\mathbf{h}}{\|\mathbf{h}\|_2}, \quad \ell = W\mathbf{h} + \mathbf{b}. \quad (4.4)$$

During evaluation, cosine similarity is calculated as the dot product between normalized query and gallery embeddings. Gallery images are ranked for each query,

and performance is reported using Rank- k and mean average precision (mAP). The model was trained for 30 epochs, and the validation-selected representation was evaluated once under the held-out Protocol A conditions.

4.2.7 Oracle Segmentation-Guided Re-Identification

The segmentation-guided Re-ID model uses the ground-truth target-cow masks released with SideView. It is therefore an oracle experiment. The non-zero part of the mask is used to define a bounding box, which is expanded by about 5%. The same coordinates are used to crop the RGB image and the mask, and both are resized to 224×224 . RGB is ImageNet-normalized, while the mask stays binary in $\{0, 1\}$. The RGB image is not multiplied by the mask. The first convolution is changed to accept four channels, and the new mask-channel filters are initialized from the mean RGB weights. The remaining learning and retrieval procedures follow the RGB Re-ID baseline.

This comparison tests whether reliable target-centered cropping and explicit foreground information help cross-setting retrieval. It does not test an automatic segmentation pipeline, and it does not separate the effect of cropping from the effect of the fourth channel. The extra 3,136 weights increase the trainable parameter count from 11,197,545 to 11,200,681.

4.3 Data Collection and Preparation

4.3.1 Data Sources and Scientific Roles

All experiments use existing datasets; no new farm data were collected for this thesis. ScienceDB is the main BCS dataset, CVB and Kaggle Beef form the main behavior dataset, and SideViewCows2026 is the main Re-ID dataset. External datasets have separate stress-test roles and are not described as evaluated unless supporting results exist. Table 4.2 summarizes these roles.

ScienceDB contains 53,566 RGB images across five BCS values from 3.25 to 4.25. Its released metadata do not contain verified biological cow IDs. Therefore, the 5,653 repaired connected burst groups are the strongest safe grouping unit available. ScienceDB evaluation is described as burst-group-disjoint, passage-disjoint, or sequence-safe. It is not described as unseen-cow evaluation.

The behavior manifest contains 2,481 CVB segments and 2,793 Beef clips. Together, they form 267 protected groups and 5,274 samples. CVB is grouped by source video, while Beef is grouped by recording-session or source-video provenance. Walking appears only in CVB. Therefore, the main behavior evaluation is source/session-group-disjoint rather than cow-disjoint, and overall accuracy by itself is not enough. SideView contains 80,260 RGB images, 80,260 matching binary masks, and 110 biological cow identities. Protocol A separates 41 cows used for representation learning from 69 cows used for evaluation. This supports an identity-disjoint interpretation for the Re-ID comparisons, unlike the BCS and behavior protocols.

Table 4.2: Primary and external dataset roles. An assigned external role does not imply that evaluation has been executed.^[E04,E05,E06,E07]

Source	Role / scale	Boundary
ScienceDB	Primary BCS; 53,566 RGB images	5,653 burst groups; no verified biological cow IDs.
CVB + Beef	Primary Behavior; 5,274 sequences/clips	Source-video/session grouped; Walking is CVB-only.
SideViewCows2026	Primary Re-ID; 80,260 image/mask pairs; 110 cows	Protocol A learning/evaluation identities are disjoint.
Ruchay 2026 / Dryad	External BCS	Planned validation; score support and modality differ.
MmCows / CBVD-5	External Behavior	Planned stress tests; label/temporal compatibility required.
BECA-L / BECA-D	External Re-ID	Planned longitudinal and population-scale stress tests.
MultiCamCows2024 / OpenCows2020	Blocked / legacy	Neither is the active primary benchmark.

4.3.2 Label Harmonization and Split Protection

ScienceDB keeps its original physical BCS values. CVB labels for resting-standing, resting-lying, grazing, drinking, and walking are mapped to the five common behavior classes. Beef contributes stand, lie, eat, and drink, while rumination is excluded. This label mapping is fixed before evaluation. SideView cow identities are used only in their assigned learning or evaluation role.

Table 4.3: Canonical BCS, Behavior, and Re-ID experimental units.^[E04,E05,E06]

Source	Partition	Examples	Groups
ScienceDB	Train	37,045	3,958 bursts
ScienceDB	Validation	8,481	850 bursts
ScienceDB	Test	8,040	845 bursts
CVB + Beef	Train	3,785	184 source/session
CVB + Beef	Validation	680	39 source/session
CVB + Beef	Test	809	44 source/session
SideView learning	Train	12,753	41 cows
SideView learning	Validation	2,683	41 cows
SideView Protocol A	Gallery (parlor)	36,811	69 cows
SideView Protocol A	Query (barn)	25,260	69 cows
SideView Protocol A	Query (snapshots)	607	63 cows

The splits protect the strongest available source of provenance. ScienceDB keeps each connected burst group inside one partition. All CVB segments from the same source video stay in one partition, and all Beef clips from the same recording session stay together. SideView Protocol A separates cow identities between learning and evaluation and also defines the acquisition setting of the gallery and queries. The implemented checks found no forbidden group or identity overlap. However, this does not prove that every possible form of leakage has been removed.

4.3.3 Perception Feasibility and Component Selection

Before choosing the perception pipeline, we compared several off-the-shelf components. The assessment separated output-return rate from actual accuracy and measured mask overlap only where ground truth was available. The results supported RT-DETR-L and SAM 2.1 as practical components. They also showed that getting a valid output from a perception model does not automatically mean that the output contains the anatomy needed by the downstream task.

Pose was not included in the first paired task configurations. A small manual review on ScienceDB showed unreliable results on rear-view images, and the available landmark set did not directly describe the intended BCS anatomy. A separate real-cattle viewpoint classifier reached 86.26% accuracy, 84.96% balanced accuracy, and 0.8573 Macro-F1 on its own held-out test of 131 images. Viewpoint was not included in the first paired task models because its downstream transfer had not been validated at that time. Camera identity was not used as a replacement for viewpoint. These statements describe the design choice for the original paired models; they do not mean that pose or viewpoint is permanently rejected.

Table 4.4: Feasibility measurements; detection and pose return rates are not accuracy.^[E11]

Candidate	Sample / measure	Result
YOLOv8s	300 images; any cow returned	223/300; median 24.5 ms
Faster R-CNN v2	Same detection measure	285/300; median 470.8 ms
RT-DETR-L	Same detection measure	283/300; median 94.3 ms
RT-DETR + SAM 2.1	100 SideView images; mask overlap	Mean IoU 0.9216; Dice 0.9530
Oracle box + SAM 2.1	Ground-truth-derived prompt	Mean IoU 0.9468; diagnostic
Pose backbones	300 images each; output return	243/300; not pose accuracy

4.4 Implementation of Selected Design

Preprocessing, spatial feature extraction, temporal sequence aggregation, and multi-task optimization were implemented using PyTorch, Torchvision, OpenCV, and the Ultralytics perception interface. Deterministic CSV manifests define all image samples, biological groups, and split partitions. Checkpoint selection was governed strictly by validation-set objectives, with held-out test sets fully isolated from training and model selection. Machine-readable manifests record all configuration parameters, per-epoch loss trajectories, checkpoint checksums, and evaluation rankings to ensure strict scientific reproducibility.

4.4.1 Monolithic Hard-Shared Multi-Task Control Architecture

The monolithic hard-shared multi-task control (denoted as E1) establishes the baseline for joint parameter sharing across the three target domains. In this architecture, all three tasks share exactly one four-channel ResNet-18 spatial feature extractor, which accounts for 11,179,648 parameters (93.74% of the total model capacity).

The model accepts task-specific input tensors whose spatial dimensions are standardized to 224×224 pixels across four channels, comprising three ImageNet-normalized RGB channels and one unnormalized binary cow mask:

$$\mathbf{X}_{\text{BCS}} \in \mathbb{R}^{B_{\text{BCS}} \times 4 \times 224 \times 224}, \quad (4.5)$$

$$\mathbf{X}_{\text{Beh}} \in \mathbb{R}^{B_{\text{Beh}} \times 8 \times 4 \times 224 \times 224}, \quad (4.6)$$

$$\mathbf{X}_{\text{ReID}} \in \mathbb{R}^{B_{\text{ReID}} \times 4 \times 224 \times 224}. \quad (4.7)$$

The shared spatial backbone maps each four-channel frame to a 512-dimensional embedding $\mathbf{h} \in \mathbb{R}^{512}$. Each task then routes this representation to its task-specific prediction head:

1. **Body Condition Scoring:** The shared feature $\mathbf{h}$ is passed to the cumulative ordinal linear head (2,052 trainable parameters), producing four independent threshold logits $g_k = \mathbf{w}_k^\top \mathbf{h} + b_k$ for $k \in \{0, 1, 2, 3\}$. The ordinal binary cross-entropy loss L_{BCS} is computed following Equation (4.2).

2. **Cattle Behavior Recognition:** Each of the 8 frames in an input video sequence passes through the identical shared four-channel ResNet-18 spatial backbone, yielding a temporal sequence of frame embeddings $\mathbf{H} \in \mathbb{R}^{B_{\text{Beh}} \times 8 \times 512}$. This sequence is processed by the temporal convolutional network (TCN) head (723,973 trainable parameters), which applies two width-three Conv1D residual blocks with 256 channels, BatchNorm, GELU, and dropout (0.2), followed by temporal adaptive average pooling to 256 dimensions and a 5-class linear classifier optimized with categorical cross-entropy loss L_{Beh} .
3. **Individual Cattle Re-Identification:** During representation training across the 41 parlor training identities, the shared 512-dimensional feature $\mathbf{h}$ is passed to a 41-class linear classifier with bias (21,033 trainable parameters) optimized using cross-entropy loss L_{ReID} . For open-set retrieval evaluation across unseen cows, unit L2-normalized embeddings $\mathbf{z} = \mathbf{h} / \|\mathbf{h}\|_2$ are extracted.

The monolithic hard-shared multi-task control contains exactly 11,926,706 trainable parameters.

4.4.2 Modular Task-Private Multi-Task Architecture

The modular task-private multi-task architecture (denoted as E3) evaluates whether introducing task-private capacity can mitigate negative transfer while retaining a shared visual backbone. To provide a controlled comparison with the hard-shared architecture, the modular architecture preserves the same shared backbone, task heads, input representations, datasets, batch sizes, losses, optimizer, training budget, and checkpoint-selection rule. It differs through the addition of three task-private residual adapters comprising 395,904 trainable parameters (+3.32% capacity). Therefore, observed differences characterize the modular configuration as a whole and cannot be attributed to routing alone. Figure 4.3 illustrates the structural comparison between the monolithic hard-shared control (which also forms the exact architectural substrate for the gradient-projected E4 model) and the modular task-private architecture (E3), including the residual bottleneck adapter design. The architectural modification consists of inserting one independent residual bottleneck adapter between the shared 512-dimensional representation $\mathbf{h}_{\text{shared}} \in \mathbb{R}^{512}$ and each respective task head. For task $t \in \{\text{BCS}, \text{Beh}, \text{ReID}\}$, the adapter consists of:

1. A linear down-projection from 512 to 128 dimensions with weights $\mathbf{W}_{\text{down}}^{(t)} \in \mathbb{R}^{128 \times 512}$ and bias $\mathbf{b}_{\text{down}}^{(t)} \in \mathbb{R}^{128}$;
2. Layer normalization over the 128-dimensional bottleneck;
3. A GELU activation function followed by dropout ($p = 0.1$);
4. A linear up-projection from 128 back to 512 dimensions with weights $\mathbf{W}_{\text{up}}^{(t)} \in \mathbb{R}^{512 \times 128}$ and bias $\mathbf{b}_{\text{up}}^{(t)} \in \mathbb{R}^{512}$.

The adapter output $\Delta \mathbf{h}_t \in \mathbb{R}^{512}$ is added residually to the shared spatial representation to produce the task-adapted feature $\mathbf{h}_t$:

$$\mathbf{h}_t = \mathbf{h}_{\text{shared}} + \Delta \mathbf{h}_t. \quad (4.8)$$

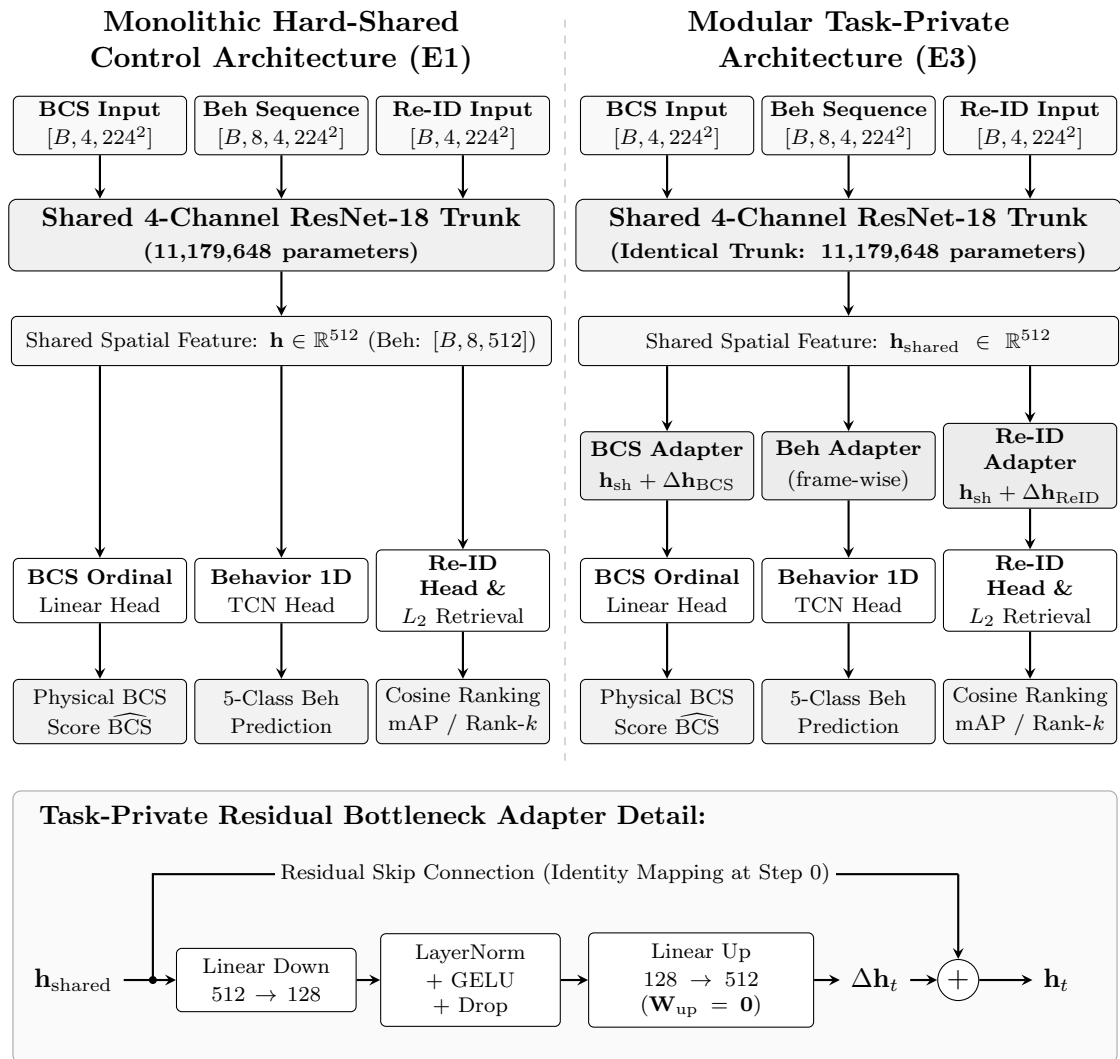


Figure 4.3: Architectural comparison between the monolithic hard-shared multi-task control (E1, left) and the modular task-private architecture (E3, right) with residual bottleneck adapters.

In the temporal behavior model, this adapter operates frame-wise on each of the 8 temporal feature vectors before the adapted sequence $\mathbf{H}_{\text{Beh}} \in \mathbb{R}^{B_{\text{Beh}} \times 8 \times 512}$ is routed to the temporal convolutional network.

Each task-private adapter contains exactly 131,968 parameters, yielding 395,904 total private parameters across the three tasks (3.21% of the model's total parameters). The modular multi-task architecture comprises 12,322,610 trainable parameters, representing an increase of 395,904 parameters (+3.32% relative capacity) over the hard-shared control.

Identity Initialization of Residual Adapters

The up-projection layer of each task adapter is initialized with zero weights and zero biases:

$$\mathbf{W}_{\text{up}}^{(t)} = \mathbf{0}, \quad \mathbf{b}_{\text{up}}^{(t)} = \mathbf{0}. \quad (4.9)$$

At step zero of training, this ensures that the adapter contribution is strictly zero:

$$\Delta \mathbf{h}_t = \mathbf{0} \implies \mathbf{h}_t = \mathbf{h}_{\text{shared}}. \quad (4.10)$$

This initialization does not imply inheriting representations from a pretrained hard-shared checkpoint; rather, it guarantees that at the start of joint training, each task-specific head receives the direct, unperturbed output of the modular model's own shared backbone. Task-private feature specialization $\Delta \mathbf{h}_t$ then develops progressively as gradient updates accumulate during training.

4.4.3 Gradient-Projected Hard-Shared Multi-Task Optimization

The third executed multi-task model, denoted as E4, evaluates whether resolving gradient conflict at the optimization level can mitigate negative transfer without introducing architectural modifications. Rather than altering the model's structural capacity, the E4 configuration preserves the exact parameterization of the monolithic hard-shared control (E1): exactly one shared four-channel ResNet-18 spatial backbone (11,179,648 parameters) and the identical three task-specific prediction heads (747,058 parameters across the BCS ordinal head, the Behavior 1D TCN, and the Re-ID classification head), yielding exactly 11,926,706 trainable parameters with zero task-private adapters and zero routing gates. All training datasets, input tensor representations (four-channel RGB plus binary cow mask at 224×224), batch sizes ($B_{\text{BCS}} = 64$, $B_{\text{Beh}} = 8$, $B_{\text{ReID}} = 32$), task loss formulations, optimizer settings, learning-rate schedules, task loss weights ($w_{\text{BCS}} = w_{\text{Beh}} = w_{\text{ReID}} = 1.0$), and validation checkpoint-selection rules are held strictly identical to E1.

The controlled experimental intervention in E4 resides exclusively in how task-specific gradients are combined on the shared backbone parameters θ_{shared} during each multi-task training super-step. When training multi-task networks with hard parameter sharing, backward passes from distinct task losses frequently produce gradient vectors that point in conflicting directions, such that the gradient update for one task increases the loss of another [23]. To address this, E4 applies Projecting Conflicting Gradients (PCGrad; Yu et al., 2020) to the shared backbone gradients.

Within each super-step, three backward passes first compute task-specific gradients on the shared backbone:

$$\mathbf{g}_{\text{BCS}} = \nabla_{\theta_{\text{shared}}} L_{\text{BCS}}, \quad \mathbf{g}_{\text{Beh}} = \nabla_{\theta_{\text{shared}}} L_{\text{Beh}}, \quad \mathbf{g}_{\text{ReID}} = \nabla_{\theta_{\text{shared}}} L_{\text{ReID}}. \quad (4.11)$$

The task gradient vectors are evaluated pairwise in a randomized order deterministically shuffled at each step using random seed 2026. For task i and comparison task j ($i, j \in \{\text{BCS}, \text{Beh}, \text{ReID}\}, i \neq j$), directional conflict is determined by the sign of their inner product. If

$$\mathbf{g}_i^\top \mathbf{g}_j < 0, \quad (4.12)$$

the two gradients point in opposing directions. The gradient vector $\mathbf{g}_i$ is then projected onto the normal plane of $\mathbf{g}_j$:

$$\mathbf{g}_i \leftarrow \mathbf{g}_i - \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_j\|_2^2 + \epsilon} \mathbf{g}_j, \quad (4.13)$$

where $\epsilon = 10^{-8}$ is a small constant for numerical stability. In accordance with the PCGrad formulation, the comparison vector $\mathbf{g}_j$ is the original, unmodified task gradient to prevent order-dependent distortion of the reference direction. If $\mathbf{g}_i^\top \mathbf{g}_j \geq 0$, no projection is applied.

Once all pairwise projections are complete, the resulting modified shared gradients $\mathbf{g}'_{\text{BCS}}$, $\mathbf{g}'_{\text{Beh}}$, and $\mathbf{g}'_{\text{ReID}}$ are summed to form the unified update vector for the shared parameters:

$$\mathbf{g}_{\text{shared}} = \mathbf{g}'_{\text{BCS}} + \mathbf{g}'_{\text{Beh}} + \mathbf{g}'_{\text{ReID}}. \quad (4.14)$$

Gradients are added together instead of being averaged in order for the experiment to remain consistent with the baseline model (E1). In the baseline model, the gradients are combined based on a weight of 1 for each task. Consequently, any deviations from the training regime can be attributed to the gradient projection method and not the learning rate.

Each task gets its unique prediction head, which is not shared between tasks. That is why, their gradients will not interfere with one another. For this reason, PCGrad is only applied to the shared ResNet-18 backbone, while the gradients of task heads remain untouched:

$$\mathbf{g}_{\text{head}}^{(\text{BCS})} = \nabla_{\theta_{\text{head}}^{(\text{BCS})}} L_{\text{BCS}}, \quad (4.15)$$

$$\mathbf{g}_{\text{head}}^{(\text{Beh})} = \nabla_{\theta_{\text{head}}^{(\text{Beh})}} L_{\text{Beh}}, \quad (4.16)$$

$$\mathbf{g}_{\text{head}}^{(\text{ReID})} = \nabla_{\theta_{\text{head}}^{(\text{ReID})}} L_{\text{ReID}}. \quad (4.17)$$

To monitor optimization dynamics throughout training without biasing model selection, the E4 training loop computes diagnostic metrics before and after projection at every super-step. For each task pair (i, j) , these include the unprojected inner product $\mathbf{g}_i^\top \mathbf{g}_j$, the unprojected cosine similarity $\cos(\mathbf{g}_i, \mathbf{g}_j) = \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_i\|_2 \|\mathbf{g}_j\|_2}$, the binary conflict indicator $\mathbb{I}(\mathbf{g}_i^\top \mathbf{g}_j < 0)$, the gradient norms before and after projection, the total number of conflict projections triggered per super-step, and the post-projection cosine similarities. These diagnostics permit direct measurement of shared-backbone gradient-direction conflict during E4 training. Table 4.5 summarizes the structural components and exact parameter counts across all three multi-task configurations.

Table 4.5: Architectural specification and parameter allocation across the monolithic hard-shared control (E1), modular task-private multi-task architecture (E3), and PCGrad hard-shared optimization control (E4). All three models share an identical four-channel ResNet-18 spatial feature extractor and identical task prediction heads. E3 introduces three independent residual bottleneck adapters (+3.32% capacity), whereas E4 preserves the exact E1 parameterization (11,926,706 parameters) and intervenes only in shared-backbone gradient projection.

Component / Specification	Hard-Shared Control (E1)	Modular Task-Private (E3)	PCGrad Hard-Shared (E4)
Input Channels	4 (RGB + Binary Mask)	4 (RGB + Binary Mask)	4 (RGB + Binary Mask)
Shared Spatial Backbone	4-channel ResNet-18	4-channel ResNet-18	4-channel ResNet-18
Shared Feature Dimension	512	512	512
BCS Prediction Head	Cumulative ordinal linear	Cumulative ordinal linear	Cumulative ordinal linear
Behavior Prediction Head	2-block residual 1D TCN	2-block residual 1D TCN	2-block residual 1D TCN
Re-ID Prediction Head	41-class linear classifier	41-class linear classifier	41-class linear classifier
Task-Private Adapters	None (direct routing)	3 residual adapters	None (direct routing)
Adapter Bottleneck Dim.	N/A	512 $\rightarrow$ 128 $\rightarrow$ 512	N/A
Optimization Distinction	Standard gradient sum	Standard gradient sum	PCGrad projection on shared gradients
Shared Backbone Parameters	11,179,648 (93.74%)	11,179,648 (90.72%)	11,179,648 (93.74%)
Task-Private Adapter Parameters	0 (0.00%)	395,904 (3.21%)	0 (0.00%)
BCS Adapter	–	131,968	–
Behavior Adapter	–	131,968	–
Re-ID Adapter	–	131,968	–
Task Prediction Head Parameters	747,058 (6.26%)	747,058 (6.06%)	747,058 (6.26%)
BCS Ordinal Head	2,052	2,052	2,052
Behavior 1D TCN Head	723,973	723,973	723,973
Re-ID Classification Head	21,033	21,033	21,033
Total Trainable Parameters	11,926,706	12,322,610	11,926,706
Capacity Difference vs E1	Baseline	+395,904 (+3.32%)	0 (0.00%)

4.4.4 Multi-Task Joint Optimization Protocol

Because cattle datasets from different farms and recording modalities do not contain overlapping annotations across body condition, behavior sequences, and individual identities, joint multi-task training cannot employ single multi-label batches. Instead, an interleaved mini-batch schedule is executed.

Each optimization step is structured as a multi-task super-step comprising three sequential forward and backward passes:

1. **BCS mini-batch:** A batch of $B_{\text{BCS}} = 64$ samples is passed through the model. The ordinal binary cross-entropy loss L_{BCS} is computed, and gradients are backpropagated into the active parameters.
2. **Behavior mini-batch:** A batch of $B_{\text{Beh}} = 8$ video sequences (64 total frames) is passed through the model. The categorical cross-entropy loss L_{Beh} is computed, and gradients are backpropagated.
3. **Re-ID mini-batch:** A batch of $B_{\text{ReID}} = 32$ images is passed through the model. The identity cross-entropy loss L_{ReID} is computed, and gradients are backpropagated.
4. **Parameter update:** In the monolithic control (E1) and modular architecture (E3), shared backbone gradients are accumulated directly by standard summation ($\mathbf{g}_{\text{shared}} = \sum_t \nabla_{\theta_{\text{shared}}} L_t$) before a single AdamW parameter update. In the PCGrad optimization control (E4), task gradients on the shared backbone are captured independently, pairwise conflicts are projected via PCGrad, and the projected gradients are summed ($\mathbf{g}_{\text{shared}} = \sum_t \mathbf{g}'_t$) before a single AdamW parameter update, with task-head gradients remaining unmodified.

Equal task weighting is enforced throughout training:

$$w_{\text{BCS}} = w_{\text{Behavior}} = w_{\text{ReID}} = 1.0. \quad (4.18)$$

Training runs for 30 epochs with 538 super-steps per epoch (16,140 total super-steps). This schedule is defined by the complete ScienceDB BCS training population ($N = 34,369$ images; batch size $B_{\text{BCS}} = 64$), where $\lceil 34,369/64 \rceil = 538$ batches per epoch establish the super-step cadence across all three tasks. Optimization uses AdamW with an initial learning rate of $\eta_0 = 10^{-4}$, weight decay of 10^{-4} , and a cosine annealing learning-rate schedule decaying to $\eta_{\text{min}} = 10^{-6}$.

Model checkpoint selection is determined strictly by minimizing the predefined mean validation loss objective:

$$L_{\text{MTL, val}} = \frac{1}{3} (L_{\text{BCS, val}} + L_{\text{Behavior, val}} + L_{\text{ReID, val}}). \quad (4.19)$$

Held-out test sets were excluded from training, hyperparameter tuning, threshold calibration, and checkpoint selection according to the implemented evaluation protocol. Table 4.6 outlines the per-task batch parameters and joint optimization hyperparameters.

Table 4.6: Multi-task training protocol and joint optimization hyperparameters. Training executes interleaved task-specific mini-batches per super-step with gradient accumulation before parameter updates.

Target Task	Training Unit	Batch Size	Task Head	Loss Formulation
Body Condition Scoring	Single image	64	Ordinal threshold linear	Ordinal binary cross-entropy
Behavior Recognition	8-frame sequence	8 seqs	Residual 1D TCN	Categorical cross-entropy
Cattle Re-Identification	Single image	32	41-class classifier	Categorical cross-entropy

Shared Multi-Task Optimization Hyperparameters	
Training Budget	30 epochs with 538 super-steps per epoch (16,140 total super-steps; $\lceil 34,369/64 \rceil = 538$)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)
Learning Rate Schedule	Initial $\eta_0 = 10^{-4}$, cosine annealing decaying to minimum $\eta_{\min} = 10^{-6}$
Task Loss Weighting	Equal weighting: $w_{\text{BCS}} = 1.0, w_{\text{Beh}} = 1.0, w_{\text{ReID}} = 1.0$
Checkpoint Selection Rule	Minimum mean validation loss: $L_{\text{MTL, val}} = \frac{1}{3}(L_{\text{BCS, val}} + L_{\text{Beh, val}} + L_{\text{ReID, val}})$
Shared Gradient Integration	E1/E3: standard accumulated shared gradients; E4: pairwise PCGrad-projected shared gradients, then summed

4.4.5 Architectural Gradient Isolation and Claim Boundaries

The mathematical distinction in gradient routing across the three executed multi-task configurations is formal and structural:

- In the **monolithic hard-shared control (E1)**, the parameter vector of the shared backbone θ_{shared} directly receives gradient contributions from all three task loss functions:

$$\mathbf{g}_{\text{shared}} = \nabla_{\theta_{\text{shared}}} L_{\text{BCS}} + \nabla_{\theta_{\text{shared}}} L_{\text{Beh}} + \nabla_{\theta_{\text{shared}}} L_{\text{ReID}}. \quad (4.20)$$

Each task head $\theta_{\text{head}}^{(t)}$ receives gradients solely from its own loss L_t .

- In the **modular task-private model (E3)**, the shared backbone θ_{shared} intentionally receives accumulated gradients from all three tasks through the residual connection:

$$\mathbf{g}_{\text{shared}} = \sum_{t \in \{\text{BCS}, \text{Beh}, \text{ReID}\}} \nabla_{\theta_{\text{shared}}} L_t. \quad (4.21)$$

In contrast, each task-private adapter $\theta_{\text{adapter}}^{(t)}$ and task head $\theta_{\text{head}}^{(t)}$ receive gradient signals strictly from their corresponding task loss L_t , with $\nabla_{\theta_{\text{adapter}}^{(t)}} L_{t'} = \mathbf{0}$ for all $t' \neq t$.

- In the **gradient-projected hard-shared control (E4)**, the architecture is identical to E1, but the shared backbone parameter vector θ_{shared} receives the sum of pairwise conflict-projected task gradients:

$$\mathbf{g}_{\text{shared}} = \mathbf{g}'_{\text{BCS}} + \mathbf{g}'_{\text{Beh}} + \mathbf{g}'_{\text{ReID}}, \quad (4.22)$$

while each task head $\theta_{\text{head}}^{(t)}$ receives its unmodified, unprojected task-specific gradient $\nabla_{\theta_{\text{head}}^{(t)}} L_t$.

This gradient isolation is an architectural routing and optimization property that prevents direct gradient updates from one task head from altering the private adapter parameters of another task, or projects conflicting shared-backbone components. It does not constitute theoretical proof that gradient conflict in the shared backbone is eliminated, nor does it guarantee that negative transfer will be prevented. The practical consequences of these routing and optimization structures on generalization across the three target domains remain an empirical question evaluated in Chapter 5. Figure 4.4 illustrates the multi-task super-step execution sequence and the resulting gradient integration pathways and isolation boundaries.

4.4.6 Evaluation Metrics and Protocol Harmonization

BCS is evaluated with physical-score MAE, exact accuracy (Acc@0), accuracy within one adjacent class or ± 0.25 BCS units (Acc@1), balanced accuracy, and Macro-F1. For N samples,

$$\text{MAE}_{\text{BCS}} = \frac{1}{N} \sum_{i=1}^N |s_i - \hat{s}_i|. \quad (4.23)$$

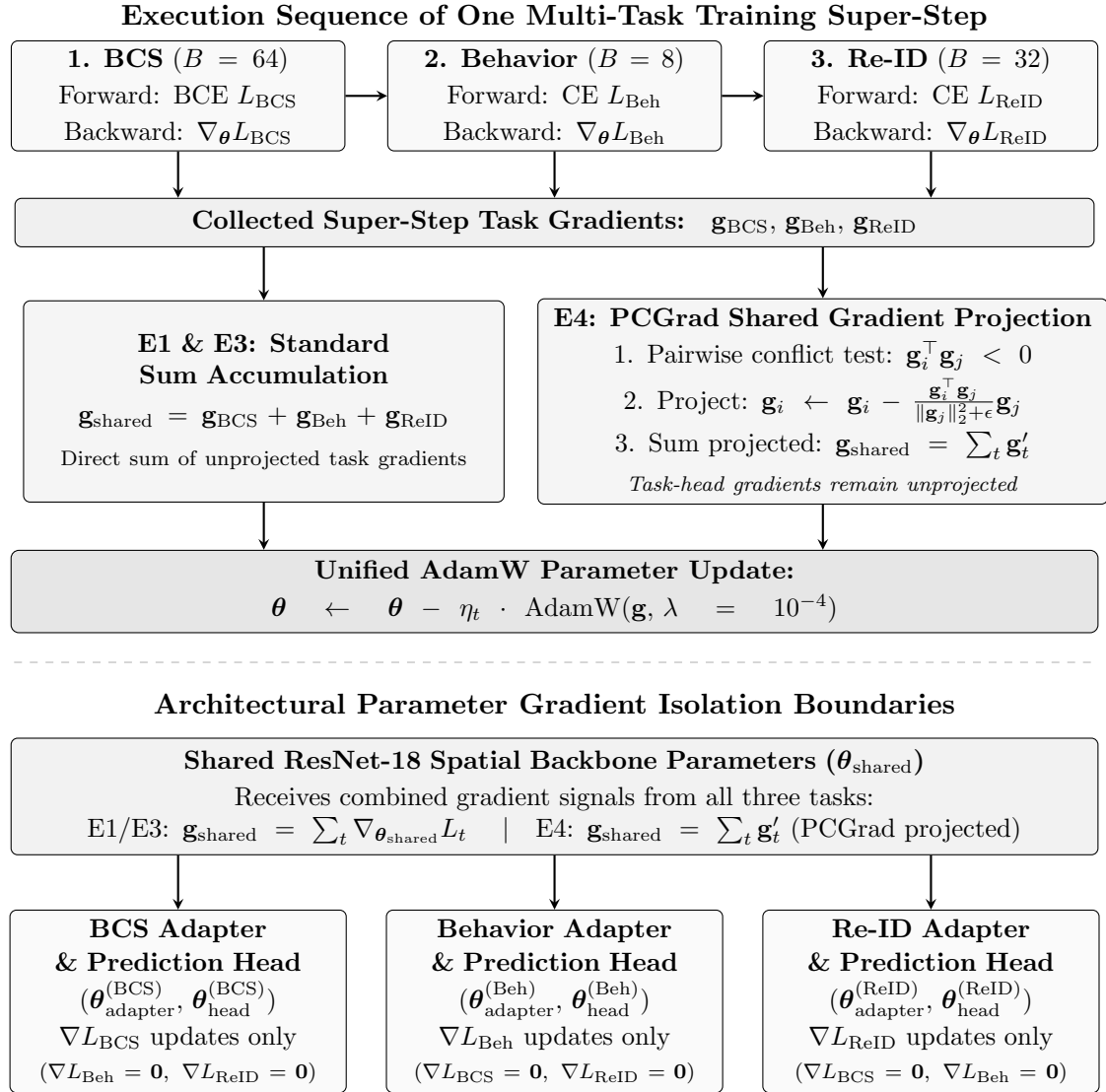


Figure 4.4: Multi-task training super-step protocol and architectural gradient routing. Tasks execute sequentially per super-step, with PCGrad (E4) projecting pairwise conflicting gradients on the shared backbone prior to parameter updates.

Behavior evaluation reports overall accuracy, balanced accuracy, Macro-F1, per-class precision, recall, and F1, and source-specific metrics. Balanced accuracy is the mean recall over the five classes, which provides a calibrated assessment across unbalanced source distributions.

For Re-ID, Rank- k measures the percentage of queries that return at least one correct identity within the top- k gallery retrieval results. Average precision summarizes ranking quality for each individual query, and mAP is the mean across all evaluated queries. Barn-to-Parlor and Snapshots-to-Parlor queries are evaluated separately against the 36,811-image parlor gallery.

To maintain strict scientific comparability, all three multi-task models—the monolithic hard-shared control (E1), the modular task-private architecture (E3), and the PCGrad optimization control (E4)—are evaluated on the exact matched held-out test populations established for the single-task reference models: 7,549 ScienceDB test images for BCS, 780 retained video sequences for behavior recognition, and 69 unseen cattle identities under SideView Protocol A (25,260 barn queries and 607 snapshot queries).

Chapter 5

Result Analysis

5.1 Performance Evaluation

The chapter will include all the finished results from single-task experiments and multi-task experiments which include Hard-Shared Baseline, Modular Task-Specific Model, and PCGrad control. Single-task experiments will first prove whether the cattle-centric representation outperforms the traditional RGB input on the same data. Multi-task experiments will then address the key issues regarding joint training. They will examine whether there is a negative transfer when only one spatial representation is being shared; whether having different task-specific modules mitigates performance loss; whether PCGrad solves any gradient conflict within the shared backbone.

5.1.1 Body Condition Scoring Results

The Baseline RGB Run 1 serves as the principal reference result for the ScienceDB images used for testing which number 8,040 in total. The physical MAE achieved by the model is equal to 0.1848 BCS units, while the Acc@1 value is 86.74%. Most of the time, the prediction error was within ± 0.25 units, while predicting the score precisely was not easy. This should be considered as one estimate since the split was disjoint only among the burst groups and not cows.^[E08]

The Perception-Enhanced model (Run 4) requires both valid bounding-box detection and non-empty segmentation. The complete detector-segmenter pipeline succeeded on 7,549 of 8,040 test images, achieving 93.89% perception coverage (489 detection misses and two empty masks). To prevent sample attrition from confounding comparisons, the baseline model was re-evaluated on this identical 7,549-image matched subset.^[E24]

Table 5.1: ScienceDB BCS results. The valid direct comparison is between the two 7,549-image matched rows; the full Run 1 result is included as the canonical reference.^[E08,E24]

Configuration	N	MAE	Acc@1	Acc@0	Bal. acc.	Macro-F1
Run 1 RGB, full test	8,040	0.1848	86.74%	41.63%	40.41%	0.4110
Run 1 RGB, matched	7,549	0.1929	84.95%	40.84%	40.23%	0.4074
Run 4 cattle-centered	7,549	0.1709	89.40%	43.57%	39.70%	0.4039

On this matched subset, the perception-enhanced model improved primary ordinal criteria (Table 5.1), lowering physical MAE from 0.1929 to 0.1709 BCS units and raising tolerance accuracy (Acc@1) from 84.95% to 89.40%. In contrast, class-balanced metrics showed no corresponding gain: balanced accuracy and Macro-F1 remained flat or slightly lower. Cattle-centered inputs thus tightened error around true score boundaries without improving discrete class-balanced discrimination. These outcomes support the utility of localized crops and binary mask guidance on perception-successful images. Because localization, cropping, and mask concatenation were introduced simultaneously, improvements cannot be attributed specifically to segmentation over tighter spatial framing. Furthermore, these findings do not describe the 6.11% of test samples where automated perception failed.

5.1.2 Behavior Recognition Results

Across the 809 canonical test sequences, the Baseline Single-Frame RGB model (Run 2) attained 88.88% overall accuracy but only 71.72% balanced accuracy, highlighting substantial class disparity. Static postures such as Lying achieved high recognition, whereas the dynamic minority Walking class was severely impaired, with only five of 26 sequences correctly classified ($F1 = 0.2174$).^[E09]

Figure 5.1 illustrates this baseline disparity, showing high recall on feeding and resting postures alongside sharp deficits on minority locomotion.

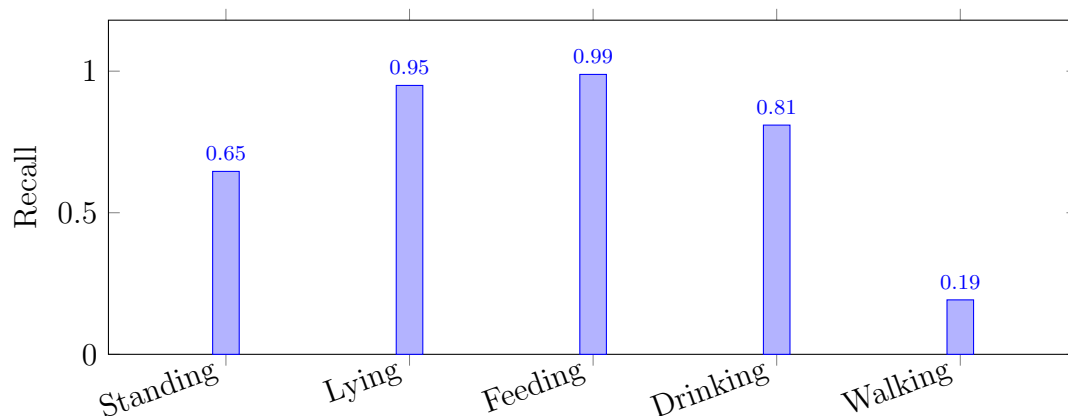


Figure 5.1: Run 2 full-test recall by class. Walking has 26 test observations and occurs only in CVB.^[E09,E12]

The perception pipeline retained 780 sequences (approximately 96.4% coverage), excluding 29 sequences due to pen stanchion occlusions. Re-evaluating Run 2 on this matched cohort establishes a fair baseline for the Perception-Enhanced Temporal model (Run 5).

As per Table 5.2, temporal foreground modeling has helped enhance balanced accuracy and test loss. While balanced accuracy went up from 71.30% to 74.43%, test loss has gone down from 0.5505 to 0.4430. However, overall accuracy was affected adversely and went down from 88.46% to 87.44%. This means that there is an improvement of the model for some classes but not necessarily in all measures.^[E27] The class-wise results (Table 5.3) clarify this trade-off: dynamic and minority categories improved—most critically lifting Walking F1 from 0.2174 to 0.2456—while dominant posture categories (Standing and Lying) experienced minor declines. The

Table 5.2: Behavior results. Direct comparison uses the same 780 retained test sequences; the 809-sample Run 2 row is the canonical full-test reference.^[E09,E27]

Configuration	N	Accuracy	Bal. acc.	Macro-F1	Loss
Run 2 RGB, full test	809	88.88%	71.72%	0.7413	0.5312
Run 2 RGB, matched	780	88.46%	71.30%	0.7378	0.5505
Run 5 perception + Conv1D	780	87.44%	74.43%	0.7397	0.4430

Table 5.3: Class-wise F1 on the matched 780-sequence Behavior test. Walking occurs only in CVB.^[E27]

Class	Run 2 RGB	Run 5 perception + Conv1D	Change
Standing	0.7556	0.7215	-0.0341
Lying	0.9512	0.9169	-0.0343
Feeding	0.9218	0.9465	+0.0247
Drinking	0.8430	0.8682	+0.0252
Walking	0.2174	0.2456	+0.0282

combined foreground-guided temporal configuration improved selected dynamic and minority-class outcomes, while some posture-dominated classes declined.

Table 5.4: Source-specific metrics on the matched Behavior test. Both sources occur in training; these are not external-transfer results.^[E27]

Source	Configuration	N	Accuracy	Bal. acc.	Macro-F1
CVB	Run 2 RGB	422	84.12%	60.62%	0.6541
CVB	Run 5 perception + Conv1D	422	80.09%	61.05%	0.6188
Beef	Run 2 RGB	358	93.58%	90.48%	0.9079
Beef	Run 5 perception + Conv1D	358	96.09%	94.26%	0.9414

Source-disjoint evaluation demonstrates pronounced domain dependence (Table 5.4). Performance on pasture cattle clips (Kaggle Beef) advanced to 96.09% accuracy and 0.9414 Macro-F1. Conversely, authentic barn surveillance (CVB) saw overall accuracy decline from 84.12% to 80.09%. Because Run 5 simultaneously integrated foreground masks and 1D temporal convolution, these shifts cannot be attributed independently to mask guidance versus sequence modeling.

5.1.3 Cattle Re-Identification Results

The open set biometric identification experiment was done by using SideView Protocol A. The system scanned through a fixed database of 36,811 images from the parlor using barn CCTV and handheld camera images. The experiment involved 69 cows that were not among the 41 training cows.^[E22,E29]

In case of Barn-to-Parlor experiments, the Oracle Segmentation-Guided model (Run 6) enhanced the main retrieval performance in Table 5.5. Accuracy for Rank-1 increased from 58.64% to 63.90%, whereas the mAP increased from 38.32% to 40.68%. It indicates that the model performed better when identifying the right

Table 5.5: Protocol A Barn-to-Parlor retrieval on 25,260 queries, 36,811 gallery images, and 69 held-out cows.^[E22,E29]

Configuration	Rank-1	Rank-5	Rank-10	mAP
Run 3 RGB	58.64%	78.19%	83.72%	38.32%
Run 6 oracle GT-guided	63.90%	77.10%	82.58%	40.68%
Change	+5.26 pp	−1.09 pp	−1.14 pp	+2.36 pp

cow among the retrieved images. But the performance on the deeper ranks was more or less similar or even worse.

Table 5.6: Protocol A Snapshot-to-Parlor retrieval on 607 queries from 63 held-out cows and a 36,811-image gallery.^[E22,E29]

Configuration	Rank-1	Rank-5	Rank-10	mAP
Run 3 RGB	38.88%	57.17%	64.58%	27.05%
Run 6 oracle GT-guided	62.93%	75.29%	81.05%	40.42%
Change	+24.05 pp	+18.12 pp	+16.47 pp	+13.37 pp

The handheld Snapshot-to-Parlor evaluation yielded a substantially larger effect (Table 5.6). Under unconstrained mobile queries, Rank-1 accuracy rose from 38.88% to 62.93% (+24.05 pp), alongside an mAP rise from 27.05% to 40.42%. The oracle target-centered configuration produced its largest improvement under the Snapshot-to-Parlor setting, where acquisition differences were greater.

These outcomes must be interpreted strictly within an oracle evaluation context. Because Run 6 uses released ground-truth masks for cropping and channel fusion, these results do not establish that automated upstream segmentation would achieve identical gains in deployment. Furthermore, target framing and mask guidance were introduced jointly, precluding isolated attribution to either factor.

5.1.4 Perception Coverage and Supporting Viewpoint Evidence

Automated upstream filtering inherently restricts evaluation scope to samples where perception succeeds. Table 5.7 summarizes retention rates across the completed evaluations.

The evaluated BCS cohort covers 93.89% of the canonical test set, while Behavior covers 96.42% of sequences (with 29 sequences excluded due to stanchion occlusion). SideView provides complete ground-truth mask coverage across all 80,260 images, representing an oracle benchmark rather than autonomous field availability.

An auxiliary three-class viewpoint classifier (front, side, rear) achieved 86.26% accuracy and 0.8573 Macro-F1 on 131 held-out images, confirming task feasibility on its own domain. However, because cross-domain feature transfer to the downstream task datasets was not independently validated, viewpoint conditioning was excluded from the primary multi-task pipeline.^[E31]

Table 5.7: Coverage of the cattle-centered inputs used in Runs 4–6. SideView coverage uses released ground-truth masks and is not an automatic-perception result.^[E24,E26,E27,E29]

Evaluation	Retained / candidate	Coverage	Exclusions or boundary
Run 4 BCS test	7,549 / 8,040	93.89%	489 detector and 2 SAM failures.
Run 5 Behavior Train+Val	4,271 / 4,465	95.65%	194 invalid-mask exclusions.
Run 5 Behavior test	780 / 809	96.42%	29 perception exclusions.
Run 6 SideView Protocol A	All gallery/query images	100%	Paired masks supplied by the dataset.

5.2 Multi-Task Learning Results

To evaluate joint optimization across the three cattle monitoring domains, all multi-task configurations were tested on the exact matched held-out populations established for single-task references:

- **Body Condition Scoring:** $N = 7,549$ ScienceDB images under the burst-group-disjoint protocol (identical-passage safe, not biological cow-disjoint).
- **Behavior Recognition:** $N = 780$ sequence samples ($T = 8$ frames; 6,240 total frames) across barn CCTV (CVB, $N = 422$) and feedlot clips (Kaggle Beef, $N = 358$), with Walking exclusive to CVB.
- **Cattle Re-Identification:** SideView Protocol A across 69 unseen cattle against the 36,811-image parlor gallery (25,260 barn queries and 607 snapshot queries) with released oracle target masks.

Checkpoint selection was governed strictly by minimizing composite validation loss, with held-out test sets remaining untouched until final one-time post-training evaluation.

5.2.1 Body Condition Scoring Multi-Task Outcomes

The Table 5.8 demonstrates that BCS performance deteriorated when hard-shared multi-task approach (E1) was applied relative to the performance of the baseline (Run 4). Physical mean absolute error became equal to 0.1709 for Run 4 and 0.1788 for E1. Other metrics like tolerance accuracy and balanced performance metric became worse as well. Ordinal cross-entropy loss became somewhat better but did not contribute to the improvement of BCS score prediction.

Neither modular adapters nor gradient projection overcame this performance deficit. Introducing task-private bottleneck adapters (E3) degraded performance further, resulting in the highest error across all models ($MAE = 0.1916$). The PCGrad Hard-Shared Optimization Control (E4) partially mitigated this loss, lowering MAE to 0.1828 and recovering tolerance accuracy relative to E3. However, E4 still trailed

Table 5.8: Multi-task Body Condition Scoring results on the matched held-out ScienceDB test population ($N = 7,549$ images). For MAE and Loss, lower values are better ($\downarrow$); for Acc@1, Acc@0, Balanced Accuracy, and Macro-F1, higher values are better ($\uparrow$). All models share the four-channel ResNet-18 spatial trunk.^[E24,E36,E37,E38]

Configuration	MAE	Acc@1	Acc@0	Bal. Acc.	Macro-F1	Loss
Single-Task Reference (Run 4)	0.1709	89.40%	43.57%	39.70%	0.4039	0.4403
Monolithic Hard-Shared (E1)	0.1788	88.44%	41.10%	35.70%	0.3605	0.4175
Modular Task-Private (E3)	0.1916	86.32%	38.47%	33.13%	0.3291	0.4555
PCGrad Hard-Shared (E4)	0.1828	87.84%	40.23%	34.97%	0.3516	0.4181

the dedicated single-task reference (MAE = 0.1709). Crucially, all three multi-task configurations underperformed the single-task model on primary physical and balanced metrics, confirming persistent negative transfer for body condition scoring under joint training.

5.2.2 Behavior Recognition Multi-Task Outcomes

Joint training under naive hard sharing (E1) produced severe degradation across balanced evaluation measures (Table 5.9). Balanced accuracy plunged from 74.43% in the single-task reference to 67.30%, accompanied by sharp declines in overall accuracy and Macro-F1. The minority Walking class was disproportionately impacted, decreasing from an F1 of 0.2456 to 0.0408. Modular adapters (E3) offered only selective, loss-specific relief; while E3 improved test loss and overall accuracy, balanced accuracy eroded further (66.70%) and Walking F1 fell to 0.0000.

Table 5.9: Multi-task Behavior Recognition results on the matched held-out test population ($N = 780$ sequences, $T = 8$ frames). Higher values are better ($\uparrow$) for accuracy and F1 metrics; lower values are better ($\downarrow$) for loss. The single-task reference is Run 5.^[E27,E36,E37,E38]

Configuration	Accuracy	Bal. Acc.	Macro-F1	Loss	Walking F1
Single-Task Reference (Run 5)	87.44%	74.43%	0.7397	0.4430	0.2456
Monolithic Hard-Shared (E1)	85.00%	67.30%	0.6866	0.6226	0.0408
Modular Task-Private (E3)	85.77%	66.70%	0.6755	0.4193	0.0000
PCGrad Hard-Shared (E4)	86.92%	69.15%	0.7114	0.5454	0.0909

The PCGrad optimization control (E4) was the most effective multi-task variant for Behavior, partially mitigating hard-sharing degradation. E4 raised balanced accuracy to 69.15% and recovered Walking F1 to 0.0909 (more than doubling E1). Moreover, on authentic CVB barn surveillance (Table 5.10), E4 reached an overall accuracy of 81.04%, marginally surpassing the single-task reference. However, source-disjoint results emphasize that overall accuracy and class-balanced fidelity diverge: CVB balanced accuracy (55.30%) and Walking F1 under E4 remained far below the dedicated single-task reference (61.05% and 0.2456, respectively). Thus, while PCGrad provided clear optimization-level mitigation, it did not fully restore single-task discriminative power on difficult minority states.

Table 5.10: Multi-task Behavior Recognition performance breakdown across acquisition sources: authentic barn surveillance (CVB, $N = 422$) and feedlot camera clips (Kaggle Beef, $N = 358$). Note that the minority Walking class is exclusive to CVB.^[E27,E36,E37,E38]

Configuration	CVB Barn CCTV ($N = 422$)			Kaggle Beef Feedlot ($N = 358$)		
	Accuracy	Bal. Acc.	Macro-F1	Accuracy	Bal. Acc.	Macro-F1
Single-Task (Run 5)	80.09%	61.05%	0.6188	96.09%	94.26%	0.9414
Hard-Shared Control (E1)	76.78%	48.90%	0.5402	94.69%	93.11%	0.9236
Modular Task-Private (E3)	80.33%	60.61%	0.6026	92.18%	87.29%	0.8849
PCGrad Hard-Shared (E4)	81.04%	55.30%	0.6141	93.85%	92.05%	0.9158

5.2.3 Cattle Re-Identification Multi-Task Outcomes

The dedicated single-task oracle model (Run 6) maintained the strongest performance across both query settings, achieving a Rank-1 accuracy of 63.90% on Barn queries and 62.93% on Snapshot queries, with mAP exceeding 40% in both environments (Table 5.11). Monolithic hard sharing (E1) degraded these principal retrieval measures considerably, dropping Barn Rank-1 to 57.38% and Snapshot Rank-1 to 57.17%, alongside substantial losses in ranking precision (mAP).

Table 5.11: Multi-task Cattle Re-Identification retrieval under SideView Protocol A on 69 unseen cattle identities evaluated against the 36,811-image fixed parlor gallery. Evaluated with released SideView target masks matching the oracle single-task reference (Run 6).^[E29,E36,E37,E38]

Configuration / Model	Rank-1 $\uparrow$	Rank-5 $\uparrow$	Rank-10 $\uparrow$	mAP $\uparrow$
<i>Barn-to-Parlor Video Queries (25,260 queries)</i>				
Single-Task Reference (Run 6)	63.90%	77.10%	82.58%	40.68%
Monolithic Hard-Shared (E1)	57.38%	73.33%	79.79%	30.37%
Modular Task-Private (E3)	49.08%	67.72%	75.03%	28.04%
PCGrad Hard-Shared (E4)	54.97%	68.81%	74.99%	30.23%
<i>Snapshots-to-Parlor Handheld Queries (607 queries)</i>				
Single-Task Reference (Run 6)	62.93%	75.29%	81.05%	40.42%
Monolithic Hard-Shared (E1)	57.17%	75.45%	82.70%	33.69%
Modular Task-Private (E3)	53.71%	72.32%	80.23%	28.78%
PCGrad Hard-Shared (E4)	57.00%	72.16%	78.25%	33.87%

Different results were obtained from the architecture and optimization approaches in Re-ID. The modular adapters (E3) approach decreased the performance to Barn Rank-1 of 49.08%. PCGrad (E4) improved the performance in comparison with the previous one, and achieved a result of Barn Rank-1 of 54.97%, and a higher value of Snapshot mAP of 33.87%. Nevertheless, its results were inconsistent with E1 and were still lower than those of the single-task network. Minor improvements occurred at higher ranks, for example, Snapshot Rank-10 was slightly increased in E3. It can be concluded that there is a difference in the retrieval performance by the implementation of multiple tasks for feature sharing; however, this method is still not as accurate as a specific single-task Re-ID approach.

5.2.4 Observed Shared-Backbone Gradient Conflict

Table 5.12 reports the gradient conflict diagnostics recorded during E4 PCGrad training across all 30 epochs (16,140 super-steps). Across this schedule, the diagnostics engine recorded 44,177 pairwise conflict projections, averaging 2.737 projections per super-step (out of a theoretical maximum of 6). Conflict frequencies were remarkably uniform across task pairings: BCS and Behavior gradients conflicted in 47.8% of super-steps, BCS and Re-ID in 46.5%, and Behavior and Re-ID in 47.4%.

Table 5.12: Observed shared-backbone gradient conflict diagnostics recorded during E4 PCGrad multi-task training across all 30 epochs (16,140 super-steps). Conflicts were evaluated pairwise on the shared four-channel ResNet-18 parameter gradients prior to projection surgery.^[E35]

Diagnostic Metric / Evaluated Task Pair	Observed Training Quantity
BCS vs. Behavior Gradient Conflict Frequency	47.8% of super-steps
BCS vs. Re-ID Gradient Conflict Frequency	46.5% of super-steps
Behavior vs. Re-ID Gradient Conflict Frequency	47.4% of super-steps
Total Executed Multi-Task Super-Steps	16,140
Total Triggered Pairwise Conflict Projections	44,177
Mean Conflict Projections per Super-Step (out of 6 max)	2.737

As per the training diagnostics, conflicting gradients are observed quite often in the four-channel ResNet-18 backbone that is shared by both tasks. Nevertheless, one has to make sure that an essential distinction between E1 and E4 is not forgotten. E1 just summed up the gradients, whereas E4 applied the PCGrad method in order to determine the conflicting gradient directions and to project them. As a result, the E4 results demonstrate that conflicts are often observed only in the case of E4 training. They cannot be used in arguing that E1 experienced an equivalent rate of conflict or that gradient interference is the only explanation for the performance difference.

5.3 Analysis of Multi-Task Design Solutions

The empirical evaluations across single-task and multi-task configurations support distinct findings regarding the three multi-task design hypotheses:

1. **Monolithic Hard Parameter Sharing (E1):** In the case of the Monolithic Hard Parameter Sharing model (E1), all three tasks used a single shared ResNet-18 spatial backbone. In doing so, worse results were obtained for BCS, Behavior, and Re-ID. These findings indicate the presence of negative transfer and that the three tasks do not necessarily have to reinforce each other. The three tasks differ in the type of information used; body shape is used in BCS, motion in Behavior, and subtle visual cues in Re-ID. However, E1 does not allow us to specify precisely what was affected. E1 is unable to provide exact information about the impact on the common characteristics.

2. **Modular Task-Private Adapters (E3):** The Modular Task-Private Adapters (E3) introduced separate adapter blocks for each task between the backbone and the prediction head, yet the problem of negative transfer was not solved consistently. While some surveillance and loss measures have become better, BCS and Re-ID measures worsened, as well as Behavior balance accuracy. In addition to this, 395,904 trainable parameters were added in E3, making an increase of model capacity by 3.32%. This is why the improvements of the performance cannot be explained solely by task-specific routing.
3. **PCGrad Optimization Control (E4):** Model (E4) PCGrad modified the optimization process without modifying the model size. The number of trainable parameters in E4 was the same as that of E1, which is 11,926,706, while there were no adapter layers in it. PCGrad could address the negative transfer problem in some aspects, like Behavior accuracy, balanced accuracy, Macro-F1, and CVB surveillance effectiveness. PCGrad did better in important results of BCS and Re-ID than E3. Nevertheless, E4 was worse than single-task models on some major criteria.

Table 5.13 synthesizes the key design decisions and empirical dispositions established across the single-task and multi-task investigations.

These outcomes demonstrate that neither simple modular adapters nor gradient projection fully mitigated negative transfer across all evaluated cattle monitoring tasks. Dedicated single-task representations remained superior on several task-specific measures, reinforcing the thesis conclusion that representation requirements are task-dependent and that multi-task integration involves metric-dependent trade-offs.

5.4 Statistical Analysis and Uncertainty Boundaries

All single-task baselines and multi-task configurations in this chapter are evaluated as single trained checkpoints (point estimates). Repeated-seed training distributions, confidence intervals, and hypothesis tests were not computed. Consequently, reported differences are descriptive point comparisons and are not characterized as statistically significant.

The matched-sample design ensures valid internal comparisons across identical test cohorts: 7,549 ScienceDB images for BCS, 780 sequences for Behavior, and SideView Protocol A for Re-ID. However, this control does not quantify seed variability or population variance, and correlated video frames or bursts must not be treated as independent biological samples.

Similarly, recorded PCGrad conflict frequencies (Table 5.12) represent descriptive training diagnostics from a single optimization trajectory. Marginal metric differences (such as E4 vs. E1 on snapshot mAP of +0.18 pp) cannot be claimed as statistically equivalent.

Table 5.13: Key design decisions established across single-task investigations and executed multi-task configurations.^[E01,E04,E06,E11,E13,E24,E27,E29,E31,E32,E33,E34,E35,E36,E37,E38]

Question / Domain	Empirical Evidence	Architecture / Method Disposition
ScienceDB grouping	Identity semantics and overlap audit	Repaired burst groups; sequence-safe evaluation.
Behavior source	Dense-video/provenance audit	CVB + Beef primary; Walking CVB-only.
Re-ID source	MultiCam access blocker	Approved SideView Protocol A benchmark.
BCS representation	Matched Run 1/4 comparison	RT-DETR crop + binary mask channel retained.
Behavior representation	Matched Run 2/5 comparison	Eight-frame RGB+mask + lightweight Conv1D retained.
Re-ID representation	Protocol A Run 3/6 comparison	Oracle GT crop+mask used as upper-bound condition.
Pose feature branch	Unreliable rear-view and lying poses	Left out of the primary baseline setup.
Viewpoint prior	Separate held-out test; transfer unverified	Kept as an independent test tool without feature fusion.
MTL sharing strategy	Held-out E1, E3, E4 evaluations	Hard sharing dropped scores across tasks compared to single-task baselines; PCGrad helped recover behavior scores.

5.5 Cross-Task Relationships and Claim Boundaries

The three evaluation domains employ incommensurable metrics with distinct scales and optimization objectives:

- BCS uses physical Mean Absolute Error in BCS units (lower is better), Acc@1 within ± 0.25 units, and exact ordinal accuracy.
- Behavior uses overall accuracy, balanced accuracy, Macro-F1 across five classes, and per-class F1-scores (higher is better).
- Re-ID uses cumulative match characteristics (Rank-1, Rank-5, Rank-10) and mean Average Precision across open-set rankings (higher is better).

Because these metrics cannot be meaningfully combined into an aggregate score, negative transfer must be evaluated within each domain against its matched single-task reference.

The evaluation cohorts also define strict generalization boundaries:

- ScienceDB evaluates unseen burst groups from automated passage sequences, not verified unseen biological cows.
- Behavior evaluates protected video recordings and sessions, not cow-disjoint herds; Walking is evaluated solely within CVB barn surveillance.
- SideView Protocol A: The test comprises 69 biologically unique cows which have been left out of the representation learning process. However, the Re-ID systems rely on the ground-truth masks which have been released. It means that the results of the tests have been obtained using an oracle segmentation framework.

The external datasets mentioned in the bigger scheme of the research, including Ruchay, Dryad, MmCows, CBVD-5, and BECA, have not yet been validated. They are scheduled for stress testing in the future. Simply by referring to these datasets, we cannot imply that the model will generalize to other datasets/environments.

5.6 Discussion of Findings

The empirical findings presented in this chapter synthesize into three overarching insights:

In fact, applying visual information about cattle is quite useful in single task tests compared to RGB pictures. Nevertheless, the advantages of such approach may differ from task to task. Localized cropping and binary mask guidance improved physical score proximity for BCS (Table 5.1) and elevated open-set retrieval under unconstrained handheld camera angles for Re-ID (Table 5.6). For Behavior, temporal foreground modeling improved class-balanced discrimination and minority-class detection while redistributing accuracy across sources (Tables 5.2–5.4). These gains demonstrate the value of isolating animal foregrounds, while underscoring that foreground guidance does not operate as an unconditional panacea across all sub-metrics.

Second, monolithic hard parameter sharing produced consistent outcome-level negative transfer across all three domains. Relative to matched single-task baselines, joint training under a single shared backbone degraded BCS physical error (Table 5.8), depressed Behavior balanced accuracy and minority walking recognition (Table 5.9), and diminished primary Re-ID retrieval precision (Table 5.11). These matched degradations do not support an assumption of automatic benefit from hard sharing across these heterogeneous tasks.

Third, neither architectural capacity expansion nor gradient projection universally resolved this cross-task interference. Adding task-private residual bottleneck adapters (E3) yielded selective loss improvements for behavior surveillance but exacerbated degradations in body condition scoring and biometric retrieval. Optimization-level gradient projection (E4) provided more effective, selective mitigation—achieving the strongest multi-task behavior metrics and recovering ground on ordinal scoring and retrieval under exact parameter parity with the hard-shared baseline. Empirical training diagnostics (Table 5.12) confirm that opposing shared-backbone gradients occurred frequently throughout joint optimization.

Nevertheless, dedicated single-task models preserved superior performance across the principal benchmarks of all three domains. Rather than demonstrating universal synergy from joint learning, the findings are consistent with task-dependent representation and optimization requirements that are not fully served by a single shared backbone. Multi-task agricultural monitoring therefore demands careful metric-aware trade-off analysis rather than naive joint consolidation.

Chapter 6

Conclusion

6.1 Summary of Findings

The main aim of this thesis was to determine the effectiveness of cattle visual representations, in addition to multi-task learning, for monitoring dairy cattle. The tasks carried out under the thesis were: body condition scoring(BCS), behaviour recognition and re-identification of the cattle. The entire procedure took place through the process of record keeping, no data leakage, and matched-population controls to make sure its reproducible and reliable.

In the single-task setting, cattle-centered representations demonstrated clear empirical utility, although their benefits were task-dependent and non-uniform across metrics:

- The BCS task used 7,549 ScienceDB test images, as detection and segmentation were correct on these images, which accounted for 93.89% of the test images. The results were better than the normal RGB model with the use of cropped cattle images and binary masks. The MAE dropped from 0.1929 to 0.1709 BCS units, while Acc@1 within ± 0.25 increased from 84.95% to 89.40%. However, exact accuracy of classification was only 43.57%. Balanced accuracy and Macro-F1 also became slightly lower, from 40.23% to 39.70% and from 0.4074 to 0.4039.
- The testing for the behavior task was conducted on 780 video sequences, 8 frames per sequence, and 96.42% of the matched data. The cropped cattle images, foreground masks and a small 1D temporal network achieved better results in several measures than the single-frame RGB model. Balanced accuracy increased from 71.30% to 74.43%, test loss decreased from 0.5505 to 0.4430, and Walking F1 improved from 0.2174 to 0.2456. But the overall accuracy decreased slightly from 88.46% to 87.44%. The method was more effective with feedlot camera data, and slightly less effective with barn CCTV data.
- The model was tested on 69 unseen cattle during the Cattle Re-Identification using SideView Protocol A experiment. First-match accuracy for both query approaches was boosted by employing target centered crops from the ground-truth mask. For the Barn-to-Parlor case, Rank-1 was boosted from 58.64% to 63.90%, and mAP from 38.32% to 40.68%. For Snapshot-to-Parlor case, Rank-1 was boosted from 38.88% to 62.93%, and mAP from 27.05% to 40.42%.

The significant improvement for snapshots indicates that this approach may prove especially useful in cases when there is drastic difference in the pose, view and lighting of the cattle. This test scenario, however, was oracle-based, as ground-truth mask was used.

In the multi-task setting, joint training across the three heterogeneous tasks revealed substantial cross-task tension:

- The **Monolithic Hard-Shared Multi-Task Control (E1)** that forced all three tasks to pass through a single four-channel ResNet-18 spatial backbone, experienced negative transfer across all three tasks compared to matched single-task baselines: BCS MAE degraded from 0.1709 to 0.1788; Behavior balanced accuracy decreased from 74.43% to 67.30% (Walking F1 decreased from 0.2456 to 0.0408); and Re-ID Barn Rank-1 decreased from 63.90% to 57.38% (mAP decreased from 40.68% to 30.37%).
- The **Modular Task-Private Multi-Task Model (E3)**, which introduced task-private residual bottleneck adapters between the shared backbone and each task head (+395,904 trainable parameters, representing a +3.32% capacity increase over E1), failed to provide general negative-transfer mitigation. While Behavior test loss (0.4193) and barn surveillance accuracy (80.33%) improved over E1, BCS performance degraded further (MAE 0.1916), Re-ID retrieval declined substantially (Barn Rank-1 49.08%, mAP 28.04%), and Walking F1 dropped to 0.0000 on the matched held-out test set.
- The **Gradient-Projected Hard-Shared Multi-Task Optimization Control (E4)** had the same number of parameters as E1 (11,926,706). However E4 used PCGrad during only the shared backbone in training [23]. It helped reduce negative transfer in some results, but the benefit was not the same for every metric. During E4 training, direct optimization diagnostics recorded 44,177 gradient conflict projections across 16,140 super-steps (averaging 2.737 projections per super-step), showing that opposing shared-backbone gradient directions occurred regularly across task pairs (46.5% to 47.8% frequency). PCGrad improved Behavior overall accuracy (86.92%), balanced accuracy (69.15%), Macro-F1 (0.7114), and Walking F1 (0.0909) over E1, and recovered principal BCS and Re-ID metrics relative to E3. However, E4 remained below dedicated single-task models on several primary metrics.

6.2 Answers to the Research Questions

The empirical findings from the single-task and multi-task investigations provide direct answers to the three research questions established in Chapter 1.

6.2.1 Research Question 1

How do cattle-centered representations affect BCS, behavior recognition, and Re-ID relative to their RGB single-task references?

Cattle-centered visual representations improved important predictive and retrieval outcomes across all three monitoring domains, but their effectiveness was task-dependent and not uniform across metrics:

- For BCS, cattle-centered cropping and foreground mask guidance reduced MAE by 0.0220 BCS units and increased Acc@1 by 4.45 percentage points on matched test samples. The improvement was strongest on ordered-error and tolerance measures, while class-balanced metrics did not improve.
- For Behavior Recognition, foreground-guided temporal processing redistributed performance across classes and camera sources: balanced accuracy (+3.13 percentage points), test loss (-19.5%), and minority Walking F1 (+0.0282) improved, but overall accuracy decreased slightly (-1.02 percentage points) and barn CCTV surveillance accuracy declined, showing that temporal foreground isolation benefited specific dynamic transitions rather than uniformly elevating all metrics.
- For Re-ID, target-centered representation guidance produced its largest gain under extreme acquisition shifts, improving Snapshot-to-Parlor Rank-1 by 24.05 percentage points and mAP by 13.37 percentage points, while producing a more modest 5.26 percentage-point Rank-1 improvement under stationary barn surveillance.

Overall, the cattle-centred representations were useful, but not as effective for all measures and did not eliminate shortcut learning between tasks. Combined processing steps also contributed to the improvements. Cropping and masking were applied to BCS and Re-ID, and temporal convolution was applied in addition to these two to Behavior. For this reason, it is not possible to conclude that the better results are only due to segmentation.

6.2.2 Research Question 2

How can temporal information and task-specific visual cues be incorporated while preserving the different requirements of condition assessment, activity recognition, and identity matching?

The results show that each task should use the visual and temporal information that best fits its needs, instead of using the same input format for all cattle monitoring tasks:

- For Body Condition Scoring, body shape is the key biological information. The use of bounding-box crops and binary foreground masks made it easier for the model to focus on the cow while retaining the biological information of body shape that was essential for BCS. This improved prediction error without using information from multiple frames.
- In the case of Behavior Recognition, body pose and movement are both important aspects, thus requiring data from multiple frames for the model. In fact, a model using 8 frames with foreground guided TCN performed better than the single-frame model.
- For Cattle Re-ID system, details of the coat patterns and skin texture are required. The use of crops and segmentation masks played an important role in this regard. The Oracle masks turned out to be extremely useful in difficult conditions where the correct segmentation of the foreground was done.

- Anatomical pose keypoints and viewpoints have been thoroughly tested. The frozen SuperAnimal pose had a very poor detection rate (45.60% in behavior sequence cases and 72.92% in lying cattle cases) and was unable to generate the rear pelvic keypoints required for BCS. For these reasons, the pose information is not incorporated into the primary downstream pipeline. The real-cattle viewpoint had an 86.26% accuracy rate on the dataset created just for it, but it has not been incorporated into the primary models since it has not been tested in other farm environments.

Thus, visual information has to be chosen based on requirements for each specific task. The body shape and morphology are needed for BCS, for behavior recognition change over time is necessary, and unique coat surface details are necessary for re-identification. Thus, it would be more useful to do that instead of using the same visual information for all the tasks.

6.2.3 Research Question 3

How does task-conditioned sharing compare with hard sharing in task performance, negative transfer, and robustness under domain change?

The comparative evaluation of multi-task configurations against matched single-task baselines yielded four definitive conclusions:

- **Outcome-Level Negative Transfer under Hard Sharing:** The hard parameter sharing method (E1) showed inferior performance compared to the matched single-task models for all three tasks. For BCS, there was an increase in the MAE metric by +0.0079. In Behavior, there was a decline in the balanced accuracy score by -7.13 percent, and Walking F1 went down from 0.2456 to 0.0408. In Re-ID, there was a decrease in Barn Rank-1 by -6.52 percent and mAP by -10.31 percent. This is outcome-level negative transfer due to different cattle vision tasks being made to share the ResNet-18 spatial backbone.
- **Evaluation of Task-Conditioned Modularity:** Incorporating task-private residual bottleneck adapters (E3) did not provide general mitigation of negative transfer. Although E3 improved Behavior test loss (0.4193 vs. 0.6226) and barn surveillance accuracy (80.33% vs. 76.78%) over E1, it degraded BCS MAE further to 0.1916, lowered Re-ID Barn Rank-1 to 49.08%, and obtained Walking F1 = 0.0000 on the matched held-out test set. Furthermore, because E3 added 395,904 trainable parameters (+3.32% capacity over E1), its performance differences cannot be attributed purely to architectural routing.
- **Evaluation of Gradient-Projected Optimization:** PCGrad optimization control (E4), which matched E1 parameter-for-parameter without additional capacity, directly confirmed that opposing shared-backbone gradient directions occurred regularly across task pairs during training (44,177 conflict projections across 16,140 super-steps; 2.737 projections per super-step; 46.5% to 47.8% pairwise conflict frequency). Projecting conflicting gradient components during optimization provided selective mitigation: Behavior overall accuracy rose to 86.92% (+1.92 percentage points vs. E1), balanced accuracy rose to 69.15% (+1.85 percentage points), Macro-F1 rose to 0.7114 (+0.0248), and Walking

F1 rose to 0.0909 (vs. 0.0408 in E1), while recovering principal BCS and Re-ID metrics relative to E3. However, E4 did not eliminate negative transfer: dedicated single-task models remained superior on several task-specific measures (BCS MAE 0.1709 vs. 0.1828 E4; Behavior balanced accuracy 74.43% vs. 69.15% E4; Walking F1 0.2456 vs. 0.0909 E4; Re-ID Barn Rank-1 63.90% vs. 54.97% E4).

- **Robustness under Domain Change:** The executed multi-task models were evaluated under source and acquisition differences within the primary protocols, such as varying camera angles in SideView (Barn vs. Snapshot queries) and differing recording environments in Behavior (CVB barn surveillance vs. Kaggle Beef feedlot footage). However, these evaluations remain confined to the executed dataset protocols; the results provide evidence under source and acquisition changes within the executed protocols, but they do not establish broad cross-dataset generalization of the final multi-task framework to unseen external commercial herds.

6.3 Contributions of the Thesis

This thesis makes six distinct, evidence-supported contributions to computer vision for precision cattle monitoring and multi-task representation learning:

1. **Leakage-Aware Evaluation Protocols:** Established verifiable, provenance-preserving benchmark protocols for three heterogeneous cattle monitoring domains: a burst-group-disjoint protocol for ScienceDB BCS to prevent identical-passage video frame leakage; a session- and source-disjoint protocol for CVB and Kaggle Beef Behavior recognition; and an identity-disjoint open-set retrieval protocol (SideView Protocol A) evaluating 69 held-out cows against a fixed 36,811-image parlor gallery.
2. **Controlled Single-Task Empirical Baselines:** Provided rigorous single-task baseline comparisons between generic RGB inputs and cattle-centered representations across matched test populations, establishing that localized cropping and foreground segmentation yield substantial, task-dependent benefits on primary metrics (e.g., reducing BCS MAE from 0.1929 to 0.1709 and increasing Re-ID Snapshot Rank-1 from 38.88% to 62.93%).
3. **Task-Specific Cattle Representation Strategy:** A task-specific approach for designing the representation of input data for a particular task was proposed and validated. Static crops of cows and masks were used for BCS in order to capture body shapes. An 8-frame sequence of cow videos was represented using a 1D temporal convolution. For the task of re-identification, important surface patterns were maintained. Limits of pose estimation and viewpoint were also explored.
4. **Benchmark for Systematic Multi-Task Sharing:** Performed a controlled experiment in multi-task settings to compare monolithic hard parameter sharing (E1), modular task-private residual adapters (E3), and gradient-projected optimization (E4) against matched single-task baselines using identical held-out test sets, providing an empirical benchmark for multi-task livestock vision.

5. **Direct Gradient Conflict Diagnostics:** Direct empirical evidence of interference among gradients of common parameters when training multiple tasks using PCGrad, with 44,177 pairs of gradient conflicts measured in 16,140 super-steps, and showing regular gradient conflicts between agricultural monitoring tasks at a frequency of 46.5–47.8%.
6. **Calibrated Insights Into Multi-Task Trade-Offs:** Established evidence showing that multi-task learning is not always advantageous for individual agriculture tasks due to shared representation, showing that the problem of negative transfer is an outcome-based one and that solutions lead to selective, metric-dependent trade-offs instead of uniformly improved performance.

6.4 Limitations

To preserve academic integrity and scientific claim boundaries, several methodological and empirical limitations of this work must be acknowledged:

1. **Absence of Verified Cow IDs in ScienceDB:** The ScienceDB dataset does not provide individual cow identifiers. Although the burst-group-disjoint protocol strictly prevents near-identical frame leakage from automated passage sequences, it cannot guarantee biological cow-disjoint evaluation.
2. **Session-Disjoint Rather than Cow-Disjoint Behavior Evaluation:** The primary Behavior dataset combines CVB barn surveillance and Kaggle Beef feedlot clips partitioned by protected video recordings and sessions rather than biologically distinct cow identities.
3. **Confounding of Minority Walking Class:** The minority Walking behavior class occurs exclusively in the CVB barn surveillance source (26 test sequences), creating an inherent confounding between camera source and activity classification.
4. **Automatic Perception Pipeline Coverage Boundaries:** The perception-enhanced BCS pipeline achieved 93.89% coverage on ScienceDB (with 489 detector and 2 segmentation failures across 8,040 test images), and the Behavior pipeline retained 96.42% of test sequences. Performance on perception-failed instances remains outside the evaluated sample distribution.
5. **Oracle Segmentation Guidance in Re-ID:** Run 6 and all multi-task Re-ID evaluations utilized released dataset ground-truth masks rather than masks generated by an automatic upstream segmentation model. While this establishes an upper-bound benchmark for foreground isolation, it does not evaluate an automated end-to-end deployment pipeline.
6. **Joint Component Attribution:** In the single-task comparisons (Runs 4–6), localization, cropping, binary masks, and temporal convolutions were evaluated in combined configurations. The individual marginal contribution of each isolated component was not ablated independently.

7. **Capacity Confounding in Modular Architecture:** The modular adapter architecture (E3) introduced 395,904 additional trainable parameters (+3.32% capacity over E1). Consequently, observed performance differences cannot be attributed purely to architectural routing in isolation from parameter scale.
8. **Boundary on Gradient Conflict Causality:** Direct gradient conflict measurements were recorded exclusively during E4 PCGrad training. These diagnostics show that opposing shared-backbone gradient directions occurred regularly during E4 training, but they do not establish that gradient conflict was the sole causal driver of E1 degradation, nor that E1 experienced identical conflict statistics.
9. **Point-Estimate Evaluations:** All reported single-task baselines and multi-task models were evaluated from single trained checkpoints. Computational constraints precluded repeated-seed training distributions, confidence intervals, or formal hypothesis testing; reported metric differences represent descriptive point comparisons.
10. **Unexecuted External Stress Tests:** Formal evaluation remained restricted to the primary benchmark datasets. Planned external stress tests on uncalibrated datasets (such as Ruchay, Dryad, MmCows, CBVD-5, and BECA) remain unexecuted roadmap items; results do not imply verified cross-domain generalization.
11. **Unverified Downstream Transfer of Pose and Viewpoint:** Anatomical pose estimation and viewpoint priors exhibited high failure rates and source shortcut risks during feasibility audits, and their transfer to downstream task datasets was not verified.
12. **Persistence of Negative Transfer:** Although there are some gains made through PCGrad, in the end, the multi-task models performed worse than single-task models on many key metrics, which demonstrates that joint multi-tasking deployment is an deliberate engineering compromise.

6.5 Future Work

Several promising research directions emerge directly from the findings and limitations of this thesis:

- **Repeated Seed Training and Uncertainty Estimation:** Future work needs to carry out multiple seed trainings in order to estimate confidence intervals, variance bounds, and statistical tests in both single and multitask scenarios.
- **End-To-End Automatic Re-ID Segmentation:** Combining an upstream automatic segmentation model (e.g., fine-tuned SAM or YOLO-seg) which uses predicted masks instead of oracle masks, studying the effect of real-world degradation due to segmentation errors.

- **Generalization via External Testing using Multiple Herds:** Testing of our trained networks using independent external test databases, such as Ruchay2026 and Dryad for BCS and MmCows or CBVD-5 for behavior recognition, to assess cross-farm generalization.
- **Longitudinal Identity Retrieval:** Stress-testing cow Re-ID over time to determine the stability of the biometric identity across different cohorts (BECA-L and BECA-D), and across different lactation stages, seasonal coat appearance and body growth.
- **Ablation Studies for Individual Perception Components:** Performing factorial ablation studies to analyze the separate contributions of bounding box localization, soft probabilistic masking, binary foreground masking and temporal length of sequences.
- **Robust Cattle Pose Estimation:** Fine-tuning animal pose models for lying, recumbent, and rear-view cattle postures to address the problem of missing keypoints, which hinders the integration of pose information into cattle behavior monitoring.
- **Advanced Multi-Task Optimization Techniques:** Exploring partial parameter sharing approaches (including branched trunk) and adaptive loss weighting techniques (including GradNorm [10]) to address the cross-task gradient conflict.

6.6 Conclusion

The thesis was successful in establishing an empirical and methodological framework for multi-task deep learning in the context of precision cattle monitoring, involving body condition scoring, behavior recognition, and cow re-identification under controlled single-task and joint multi-task settings.

Evidence from experiments has shown that cow-focused visual representations offer significant utility in the performance of specific individual livestock monitoring tasks by improving BCS error tolerance and improving the re-identification of cows when there are unconstrained shifts in the cameras. This is however task dependent and does not address the separate needs of representation for morphological regression, temporal modeling, and biometric identification.

Naive hard parameter sharing in a multi-task integration framework led to negative transfer effects at the outcome level in all three tasks. Adding task-private residual adapters did not help mitigate the performance drop, whereas PCGrad explicitly showed that gradient directions for shared backbones were often opposite in training. Through the balancing of opposing gradient components, PCGrad delivered a selective remedy: especially improving behavior classification metrics and authentic barn surveillance performance, and yet specific single-task models were better on some task-specific metrics.

In the end, the results show that multi-task learning in agricultural computer vision is not universally beneficial in terms of performance using shared representation; successful deployment should entail a balance between shared characteristics of cattle, task-oriented abilities, and optimization strategies that account for task conflicts.